

Midpoint convexity from bounded tree potentials and path costs

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Abstract

Four real Banach spaces defined by bounded tree potentials satisfy the averaged asymptotic midpoint bound $\widehat{\delta}(t) \geq \sqrt{1+t^2/4} - 1$ for $0 < t < 1$, while none admits an asymptotically uniformly convex (AUC) renorming. On finite-height trees, the globally constrained norm equals the least additive cost of a Hilbert vector and root paths. The quadratic path and segment-start outer Hilbert sums are reflexive and asymptotically midpoint uniformly convex, and admit no equivalent AUC norm.

1 Introduction

Bounds on scalar sums along tree paths can have two different geometric effects. They keep path vectors bounded. When arbitrarily tall trees of these vectors have weakly null child increments bounded away from zero, that boundedness obstructs one-sided asymptotic convexity under every equivalent norm. At the same time, quadratic restrictions on the coefficients of a test permit a uniform gain when the two sides of a midpoint are considered together. We study explicit norms for which both effects can be calculated.

For an infinite-dimensional real Banach space X with norm N , write $\text{cof}(X)$ for the closed linear subspaces of finite codimension. The averaged midpoint and one-sided moduli used here are

$$\widehat{\delta}_N(t) = \inf_{N(x)=1} \sup_{F \in \text{cof}(X)} \inf_{\substack{y \in F \\ N(y) \geq 1}} \left(\frac{N(x+ty) + N(x-ty)}{2} - 1 \right), \quad (1)$$

$$\bar{\delta}_N(t) = \inf_{N(x)=1} \sup_{F \in \text{cof}(X)} \inf_{\substack{y \in F \\ N(y)=1}} (N(x+ty) - 1), \quad t > 0. \quad (2)$$

Asymptotic uniform convexity, abbreviated AUC, means that the second modulus is positive at every positive radius. The first requires a gain in the average of two endpoint norms. Its infimum over directions of norm at least one agrees with the infimum over unit directions: the average on each line is an even convex function of the radius. We retain the larger-direction convention because it is convenient for the quantitative estimates below.

Dilworth, Kutzarova, Randrianarivony, Revalski, and Zhivkov introduced asymptotic midpoint uniform convexity using the maximum of the two endpoint norms. They gave an AMUC norm on ℓ_2 which is not AUC, and asked whether the two properties agree after renorming [5, Definition 2.2, Theorem 2.4, and Section 5]. The maximum and averaged conventions have the same positivity property. Indeed, after intersecting a subspace with the kernel of a functional norming the center, both endpoint norms are at least one; the average excess is then at least half the maximum excess. Their numerical moduli must nevertheless be distinguished.

The example of Dilworth et al. has unit ball equal to the closed convex hull of the Hilbert unit ball and $\{\pm(e_1 + e_n) : n \geq 2\}$. Its dual description combines a Hilbert bound with bounds on the

shared-root sums of coefficients. Their proof of Theorem 2.4 selects a positive or negative part of a Hilbert tail and perturbs a dual norming functional inward [5, Lemma 2.5 and proof of Theorem 2.4]. The finite-height global models below extend the shared-root path pattern through many successive generations. The paired relative-potential argument uses the related positive-part mechanism under each of its stated quadratic constraints, and uses both inward parts.

For separable reflexive spaces with unconditional asymptotic structure, Baudier, Causey, Dilworth, Kutzarova, Randrianarivony, Schlumprecht, and Zhang proved equivalence of AMUC and AUC renormability [3, Corollary 5.4]. Perreau recorded in 2021 that the reflexive question without that structural hypothesis remained open [10, Introduction, after Theorem C]. Baudier later answered the general renorming question negatively using the Kadets–Werner modification of the Bourgain–Rosenthal construction [2, Theorem 1 and Corollary 1]. The Kadets–Werner example is infinite dimensional and has the Schur property, so it is nonreflexive [9, Theorem 2.5 and the paragraph before Corollary 2.6]. The original martingale construction is due to Bourgain and Rosenthal [4]; the Kadets–Werner modification is constructed in Section 2 of their preprint [9].

Our purpose is to give explicit tree models and to relate their midpoint geometry to the admissibility of scalar tests and the cost of path decompositions. Classical tree spaces provide further context. James defined the disjoint-segment norm on an infinite binary tree [8, p. 739], and Girardi proved AUC for the canonical predual and full dual of the James tree space with their natural norms [7, Theorems 3 and 5]. The segment tests below impose nodewise incomparability across distinct segments, a stronger condition than disjointness; their relation to earlier tree constructions is discussed when those tests are defined.

The weakly-null-tree viewpoint was motivated by Perreau’s discussion of asymptotic convexity [10, §3.3]. This is motivation for the obstruction, not an import of the norms, four-piece construction, or quantitative constants developed here.

Bounded potentials on the infinite tree

Our first four spaces have coordinates on the countably branching tree $T = \mathbb{N}^{<\omega}$, with or without a root coordinate. A coefficient family f defines the scalar potential $P_f(s) = \sum_{u \preceq s} f_u$, summing over the coordinates present in the model. We require $|P_f| \leq 1$. The additional square-sum constraint concerns either each sibling family, each finite antichain, or the whole coefficient family. Section 2 gives all four norming sets before defining their completions. Theorem 2.1 proves

$$\hat{\delta}(t) \geq \sqrt{1 + t^2/4} - 1 \quad (0 < t < 1)$$

for each of them, and rules out an equivalent AUC norm in every case. The estimate has a common proof, without identifying the four norms.

The main difficulty is that a head test can already reach the boundary of the allowed interval $[-1, 1]$. An arbitrary tail perturbation can leave that interval. We split its relative potential into two parts directed inward from the inherited head value. Taking a positive part is 1-Lipschitz, so the increments of each part are dominated by the original test coefficients. Quadratic budgets can then be combined on disjoint head and tail supports. Pairing these two tests with the two endpoints gives the estimate. Section 3 develops the different tests obtained by selecting one part, grouping parts by attachment sign, or choosing exit components separately. These are independent choices of witnesses: the later models can be read after Section 2, without passing through Section 3. The bounded-tree obstruction is proved locally in Lemma 2.3 and instantiated with each model’s actual path and child-increment bounds.

Section 4 uses edge coordinates and a free root potential, with no root coordinate contributing to evaluation. It gives a direct construction of edge witnesses: inward scalar jumps detect the first

edges exiting a head, while a contractive map into the Euclidean plane preserves the head and detects the deeper tail. The section records the bound supplied by this construction and proves its one-sided renorming obstruction directly.

Path and segment costs on finite-height trees

For $n \geq 1$, restrict the globally constrained model to $T_n = \mathbb{N}^{\leq n}$ and denote its norm by L_n . Write $b_s = \sum_{r \preceq s} e_r$ for a root-path vector and $B_n \mu = \sum_s \mu_s b_s$ for $\mu \in \ell_1(T_n)$. For $x \in \ell_2(T_n)$, Proposition 5.1 proves the exact identity

$$L_n(x) = \inf_{x=h+B_n\mu} (\|h\|_2 + \|\mu\|_1).$$

At height one this is exactly the norm of Dilworth et al. described above: the nontrivial root paths are $e_o + e_s$ for children s , and the root singleton already belongs to the Hilbert ball. This connects the potential argument with path decompositions over arbitrary heights. Section 5 examines the outer Hilbert sum of these finite-height blocks. Dual clipping controls midpoint tails; removing signed path mass gives complementary deficit estimates. Section 6 gives two estimates for dyadic martingales whose values are supported in finite heads, each chosen before the sign producing the corresponding new value. They bound the sum of expected squared norms on successive new coordinate layers, with constants 495 and 252 in Theorems 6.1 and 6.3.

Replacing the sum of costs by their Euclidean combination changes the norm. The root singleton then has norm $1/\sqrt{2}$, and the dual constraint is a sum of two squares. Section 7 proves the corresponding cancellation and stopping estimates. In particular, Theorem 7.6 bounds the tail of an arbitrary displacement by $8\sqrt{R^2 - \|x\|^2}$ when the center is supported in the head and both endpoint norms are at most R . A different stopping argument improves the coefficient to 7 in Proposition 7.10.

Finally, replacing root paths by segments introduces their starting vertices into the cost. Sections 8 and 9 analyze a quotient cost computed from the mass of segment starts and the dual of an incomparable-segment test norm, respectively. The start-mass cost gives the squared midpoint inequality with coefficient $1/64$ in Theorem 8.4. Write P_E for coordinate restriction to E and $Q_E = I - P_E$. For the incomparable-segment space, Theorem 9.5 allows an arbitrary displacement y : if $P_E x = x$ for a finite ancestral head E and $\|x \pm y\|_I \leq 1$, then

$$\|Q_E y\|_I \leq (\sqrt{2} + \sqrt{6})\sqrt{1 - \|x\|_I}.$$

This estimate comes from shortening segments with opposite boundary coefficients and controlling an unconditional mixture of the endpoint atoms. The quadratic path space X_q and the segment-start space X_S are reflexive, have positive maximum asymptotic midpoint modulus at every positive radius, and admit no equivalent AUC norm.

2 Four potential norms, one pair of inward tests

We will construct four spaces satisfying the same midpoint bound. The path constraint will keep their path vectors bounded; the quadratic constraints will permit the same pair of inward perturbations.

Let $T = \mathbb{N}^{<\omega}$, with root o , and let $T^+ = T \setminus \{o\}$. Write $r \preceq s$ when r is a predecessor of s , write s^- for the parent of a nonroot node s , and write $\text{ch}(r)$ for the children of r . On either $I = T$ or $I = T^+$, the potential of a coefficient family f is

$$P_f(s) = \sum_{u \in I, u \preceq s} f_u.$$

There are four norming sets. Their common condition is $|P_f(s)| \leq 1$ at every node; their other conditions are as follows:

K	I	support condition	quadratic condition
K_s	T^+	none	$\sum_{u \in \text{ch}(r)} f_u ^2 \leq 1 \quad (r \in T)$
K_p	T	none	$\sum_{u \in \text{ch}(r)} f_u ^2 \leq 1 \quad (r \in T)$
K_a	T	$f \in c_{00}(T)$	$\sum_{u \in A} f_u ^2 \leq 1 \quad (A \text{ a finite antichain})$
K_g	T	$f \in \ell_2(T)$	$\sum_{u \in T} f_u ^2 \leq 1.$

An antichain has no two comparable nodes. The root coefficient is present in K_p , even though it belongs to no sibling group. Its absolute value is bounded by the path condition at o .

The root-inclusive sibling constraint also has a useful potential description. A family $b : T \rightarrow [-1, 1]$ with $\sum_{u \in \text{ch}(r)} |b_u - b_r|^2 \leq 1$ determines exactly one $f \in K_p$ by $f_o = b_o$ and $f_s = b_s - b_{s-}$ for $s \neq o$. Conversely $b = P_f$ recovers every member of K_p . Thus the root label b_o is free in $[-1, 1]$; the root coefficient records that label relative to a dummy parent whose value is fixed at zero. Section 4 omits this root coordinate and studies the resulting norm on the edge coordinates. This convention differs from K_s , whose root potential is fixed at zero.

For each row, put $\|x\|_K = \sup_{f \in K} \sum_{s \in I} f_s x_s$ on $c_{00}(I)$ and denote the completion by X_K . Each test set is symmetric, its coefficients have absolute value at most one, and each signed singleton is a test. Thus this formula defines a norm, $\|e_s\|_K = 1$, and coordinate evaluations are continuous. Each test extends to the completion with norm at most one. Moreover these extensions remain norming; the supremum of their evaluations is a 1-Lipschitz function which agrees with the norm on a dense subspace. We will write $f(x)$ for the extended evaluation.

The global constraint gives the additional comparison $\|z\|_{K_g} \leq \|z\|_2$ for every finite array z . Consequently the restriction of any $\varphi \in X_{K_g}^*$ to the finite arrays is Hilbert-continuous. Its coordinate values form an $\ell_2(T)$ family. Thus every sequence of distinct coordinate vectors, and in particular every sibling sequence, is weakly null in X_{K_g} .

Theorem 2.1. *Each of the four completed spaces X_K is a separable infinite-dimensional real Banach space. For $K \in \{K_s, K_p, K_a, K_g\}$ and $0 < t < 1$,*

$$\widehat{\delta}_{X_K}(t) \geq \sqrt{1 + t^2/4} - 1. \quad (3)$$

None of these spaces admits an equivalent AUC norm.

The proof has two parts. We first obtain the midpoint bound by perturbing a norming test outside a finite set of coordinates. We then construct bounded paths whose child increments prevent AUC renorming.

A finite set $D \subset T$ is *initial* if it contains the root and every predecessor of each of its nodes. Restricting a test to D preserves every displayed condition: quadratic sums decrease, while a path sum freezes when the path leaves D . Consequently, if x is supported in $D \cap I$, there is a test f supported there with $f(x) = \|x\|_K$. Indeed we can take the supremum over tests on this fixed finite set, which form a closed bounded set in a finite-dimensional space. The tail space

$$F_D = \{y \in X_K : y_s = 0 \text{ for } s \in D \cap I\}$$

is closed and has finite codimension. Its finite vectors are dense: subtract the finitely many head coordinates from any finite approximation to $y \in F_D$.

A one-generation positive-part perturbation appears in the proof of Theorem 2.4 of Dilworth et al. [5]: they select a positive or negative part of a Hilbert tail and orient the dual perturbation inward from the root value. The next lemma clips a relative potential on each exit component and pairs both inward parts.

Lemma 2.2 (Paired inward tests). *For any of the four norms, any x supported in a finite initial D , any $y \in F_D$, and $t > 0$, one has*

$$\frac{\|x + ty\|_K + \|x - ty\|_K}{2} \geq \sqrt{1 - \theta^2} \|x\|_K + \frac{t\theta}{2} \|y\|_K \quad (0 \leq \theta \leq \tfrac{1}{2}). \quad (4)$$

Proof. Choose a head test f with $f(x) = \|x\|_K$ and an arbitrary test g . We construct two inward increment families from g . On each component of $T \setminus D$, let r be its last ancestor in D , put $c = P_f(r)$, and use the relative potential

$$R(s) = P_g(s) - P_g(r), \quad |R(s)| \leq 2.$$

Let $\sigma = -1$ if $c \geq 0$ and $\sigma = 1$ if $c < 0$, and write $\xi_+ = \max\{\xi, 0\}$ for $\xi \in \mathbb{R}$. Define

$$U(s) = \sigma(\sigma R(s))_+, \quad V(s) = \sigma(-\sigma R(s))_+,$$

and set $U = V = 0$ on D . Thus both U and V point inward from c , and $U - V = R$. Let u, v be their edge increments, with root coefficient zero if the root is a coordinate.

Taking a signed positive part is 1-Lipschitz and fixes zero. Along an edge within a component the reference ancestor is unchanged; on an edge leaving D its parent has relative potential zero. It follows in both cases that

$$|u_s| \leq |g_s|, \quad |v_s| \leq |g_s| \quad (s \notin D), \quad u_s - v_s = g_s \quad (s \notin D).$$

Each quadratic constraint on u or v is therefore at most one, using the same coordinate groups as in the chosen row of the table. Their potentials are bounded by two, so $u/2, v/2 \in K$. Thus u and v extend to continuous functionals on X_K with norm at most two. Coefficient domination also preserves finite support for K_a and square summability for K_g . The coefficient identity gives $(u - v)(y) = g(y)$ first on finite tails and then, by continuity, on F_D .

Put $a = \sqrt{1 - \theta^2}$. We claim $af + \theta u, af + \theta v \in K$. On a tail component their potentials are $ac + \theta U(s)$ and $ac + \theta V(s)$. If $c \geq 0$, each lies between $ac - 2\theta$ and ac ; if $c < 0$, each lies between ac and $ac + 2\theta$. Both intervals lie in $[-1, 1]$ because $a \leq 1$ and $2\theta \leq 1$. On D the potentials are simply those of af . For any quadratic group B , the head and tail supports are disjoint, so

$$\sum_{s \in B} |af_s + \theta u_s|^2 = a^2 \sum_{s \in B \cap D} |f_s|^2 + \theta^2 \sum_{s \in B \setminus D} |u_s|^2 \leq a^2 + \theta^2 = 1.$$

The same calculation applies to v ; all support requirements remain satisfied. This proves the claim for all four rows without identifying any two of their norming sets.

Evaluate $x + ty$ against $af + \theta u$, and $x - ty$ against $af + \theta v$. Since f vanishes on tails and u, v vanish on the head, adding gives

$$\frac{\|x + ty\|_K + \|x - ty\|_K}{2} \geq a\|x\|_K + \frac{t\theta}{2}(u - v)(y) = a\|x\|_K + \frac{t\theta}{2}g(y).$$

Take the supremum over $g \in K$ to obtain (4). $\square$

The main gain from pairing is that neither endpoint must be chosen in advance. A tail potential which is unfavorable for one endpoint supplies an inward test for the other. The argument does not lose the tail norm by first choosing one of the two clipped pieces.

Proof of the midpoint estimate. Fix $t \in (0, 1)$. For a finite unit center x and any $y \in F_D$ with $\|y\|_K \geq 1$, choose $\theta = t/\sqrt{4+t^2} < 1/2$ in Lemma 2.2. Its right side becomes $\sqrt{1+t^2/4}$. Now let x_0 be any unit vector in the completion. For every $\varepsilon > 0$, choose a finite unit vector x with $\|x - x_0\|_K < \varepsilon$, then choose D containing its support. The triangle inequality loses at most ε in the average, uniformly over all $y \in F_D$ of norm at least one. Therefore

$$\sup_{F \in \text{cof}(X_K)} \inf_{\substack{y \in F \\ \|y\|_K \geq 1}} \left(\frac{\|x_0 + ty\|_K + \|x_0 - ty\|_K}{2} - 1 \right) \geq \sqrt{1+t^2/4} - 1 - \varepsilon.$$

The subspace is allowed to depend on ε . Letting $\varepsilon \downarrow 0$ and then taking the infimum over x_0 proves (3). $\square$

The common bound is quadratic: for $0 < t < 1$,

$$\sqrt{1+t^2/4} - 1 = \frac{t^2}{4(\sqrt{1+t^2/4} + 1)} \geq \frac{t^2}{16}.$$

It follows from the same admissibility argument for all four test sets; no equality of their norms or optimality of their moduli is asserted.

2.1 Why every equivalent norm still has bounded branches

It remains to rule out every equivalent AUC norm. The next lemma uses the one-sided modulus defined in (2). Its mechanism is simple: an AUC norm would force a fixed multiplicative increase at every step along a suitably chosen path.

Lemma 2.3. *Let X be a real Banach space. Suppose $0 < m \leq M < \infty$ and $c > 0$, and, for every integer $h \geq 1$, X contains a tree $(p_s)_{s \in \mathbb{N}^{\leq h}}$ with $m \leq \|p_s\| \leq M$. For every nonterminal s , suppose the child increments $d_{s,n} = p_{s \frown n} - p_s$ are weakly null as $n \rightarrow \infty$ and $\|d_{s,n}\| \geq c$. If N is an equivalent norm with $\alpha\|\cdot\| \leq N \leq \beta\|\cdot\|$ for constants $0 < \alpha \leq \beta < \infty$, then $\bar{\delta}_N(\alpha c/(2\beta M)) = 0$.*

Proof. Write $t_0 = \alpha c/(2\beta M)$ and suppose $0 < \gamma < \bar{\delta}_N(t_0)$. At any current node p_s , the definition supplies a finite-codimensional F such that

$$N(p_s + w) \geq (1 + \gamma)N(p_s) \quad (w \in F, N(w) = t_0 N(p_s)).$$

The same inequality holds when $N(w) \geq t_0 N(p_s)$. To see this, write a shorter point on the ray as a convex combination of p_s and $p_s + w$; convexity gives $N(p_s + rw) - N(p_s) \geq r[N(p_s + w) - N(p_s)]$ for $r \geq 1$.

The quotient X/F is finite dimensional. Since $d_{s,n}$ is weakly null, choose $w_n \in F$ with $N(w_n - d_{s,n}) \rightarrow 0$. Eventually $N(w_n) \geq \alpha c/2 \geq t_0 N(p_s)$. Choosing the approximation error at most $\gamma N(p_s)/2$ gives a child satisfying

$$N(p_{s \frown n}) \geq (1 + \gamma/2)N(p_s).$$

Choose h large enough that $\alpha m(1 + \gamma/2)^h > \beta M$, and perform the selection in a tree of height h . Repeating it h times yields a path whose last vector has N -norm at least $\alpha m(1 + \gamma/2)^h$. This exceeds the uniform bound βM , a contradiction. Finally, $\bar{\delta}_N(t_0) \geq 0$: at a unit center intersect any candidate subspace with the kernel of a norming functional, on which the perturbed norms are at least one. This proves equality to zero. $\square$

Completion of Theorem 2.1. The coordinate vectors are linearly independent, so each space is infinite dimensional. Finite rational coordinate combinations are dense, so each space is separable. For $s \in I$, set $p_s = \sum_{u \in I, u \preceq s} e_u$. Its norm is one: the path condition gives the upper bound, and a singleton test gives the lower bound. In the rootless model begin at any fixed nonroot node v and use the subtree $(p_{v \frown s})_s$ so that the initial vector is nonzero. In all four models,

$$\left\| \sum_{n \in A} a_n e_{s \frown n} \right\|_K \leq \left(\sum_{n \in A} |a_n|^2 \right)^{1/2} \quad (A \subset \mathbb{N} \text{ finite}).$$

This is the sibling constraint in the first two models. Siblings form an antichain in the third and are part of the globally constrained family in the fourth. If $\phi \in X_K^*$, use $a_n = \phi(e_{s \frown n})$ in this inequality to obtain

$$\sum_{n \in A} |\phi(e_{s \frown n})|^2 \leq \|\phi\|^2.$$

Thus the child increments are weakly null, and each has norm one. Lemma 2.3 applies with $m = M = c = 1$. In particular, for every equivalent norm as in that lemma, $\bar{\delta}_N(\alpha/(2\beta)) = 0$. $\square$

3 Selecting tail tests

The paired argument uses both parts of every relative potential. Sometimes one needs a single test, or wants to decide separately which exit components contribute to each endpoint. We give these choices explicitly. They preserve more information about the witnessing test, although their numerical midpoint bounds are weaker than Theorem 2.1. The later path and segment arguments do not depend on this section; the numerical estimates below illustrate what each selection of a witness retains.

Throughout this section, K is one of the four norming sets of Section 2, with coordinate set I . Let D be a finite initial set and let $f \in K$ be supported in $D \cap I$. An exit component is a subtree whose first vertex lies outside D and whose parent r lies in D . Its inherited head value is $c = P_f(r)$. Given an arbitrary test $g \in K$, define its relative potential on that component by $R = P_g - P_g(r)$. Thus $|R| \leq 2$. The test g is chosen to evaluate a vector $y \in F_D$; it is not assumed to be supported outside D .

We first reuse the inward increment families u, v from Lemma 2.2. That proof establishes $u/2, v/2 \in K$, so their evaluations are continuous on X_K , and extends $(u - v)(y) = g(y)$ from finite tails to every $y \in F_D$.

3.1 One of two inward parts

Let u, v be the inward increments constructed in that proof. In addition to the tests used there, for $0 \leq \lambda \leq 1/2$ we have

$$\frac{f + \lambda u}{\sqrt{1 + \lambda^2}}, \quad \frac{f + \lambda v}{\sqrt{1 + \lambda^2}} \in K. \quad (5)$$

Indeed $c + \lambda U$ and $c + \lambda V$ already lie in $[-1, 1]$. Each quadratic sum is at most $1 + \lambda^2$, because the supports of the head and tail increments are disjoint. Dividing enforces the quadratic bound and preserves the path bound and support requirements.

The identity $(u - v)(y) = g(y)$ shows that one of these two inward parts retains at least half the evaluation:

$$\max\{|u(y)|, |v(y)|\} \geq |g(y)|/2 \quad (y \in F_D).$$

This is the single-part choice, available in all four models. To see two quantitative specializations, suppose $\|x\|_K = 1$, x is supported in D , $f(x) = 1$, $y \in F_D$, and $\|y\|_K \geq 1$. Evaluation of (5) at the favorable endpoint gives

$$\max_{\pm} \|x \pm ty\|_K \geq \frac{1 + \lambda t |g(y)|/2}{\sqrt{1 + \lambda^2}}.$$

For $0 < t < 1$, set $\lambda = t/2$ and take the supremum over g . The maximum is at least $\sqrt{1 + t^2/4}$; each endpoint is at least one by the head test. Thus this selection yields average gain $D(t) = (\sqrt{1 + t^2/4} - 1)/2$ at finite centers. Approximation of a unit center by a finite unit center within $D(t)/2$ gives the gain $D(t)/2$. This proves the rootless sibling estimate directly using a single chosen tail part.

The square-root normalization can also be replaced by the test $\sqrt{1 - \theta^2}f + \theta q$, where q is a chosen member of $\{u, v\}$. If $|g(y)| \geq \|y\|_K/2$, then $|q(y)| \geq \|y\|_K/4$. Taking $\theta = t/8$ gives

$$\max_{\pm} \|x \pm ty\|_K \geq \sqrt{1 - \theta^2} + t\theta \|y\|_K/4 \geq 1 + t^2/64.$$

Both endpoints remain at least one. In the globally constrained space this gives finite-center average gain $t^2/128$, and approximation within $t^2/256$ gives the completed-space gain $t^2/256$. The continuity established above applies also when the global test has infinite support.

3.2 Four pieces classified by attachment sign

For each $i \in \{-1, 1\}$, consider those exit components with $i = 1$ when $c \geq 0$ and $i = -1$ otherwise. On these components take the potentials R_+ and $(-R)_+$, and take zero on all other components and on D . This produces four nonnegative potentials $G_{i,+}, G_{i,-}$, all bounded by two. Their increments $g_{i,+}, g_{i,-}$ satisfy coordinatewise

$$|g_{i,\pm}(s)| \leq |g(s)|, \quad g \mathbf{1}_{T \setminus D} = \sum_{i=\pm 1} (g_{i,+} - g_{i,-}).$$

The coefficient bounds hold also on exit edges because the relative potential there starts at zero. Consequently $g_{i,\pm}/2 \in K$. The displayed coefficient identity holds on finite tails and extends to every $y \in F_D$ by continuity. One of the four pieces therefore has evaluation of magnitude at least $|g(y)|/4$. In particular, when $|g(y)| \geq \|y\|_K/2$, it retains at least $\|y\|_K/8$.

This four-piece choice is also available in all four models. In the root-inclusive sibling model, it gives a direct argument at an arbitrary unit center x . Fix $0 < t < 1$, put

$$\lambda = t/16, \quad a = \sqrt{1 - \lambda^2}, \quad \eta = t^2/2048,$$

and choose a finite x_0 with $\|x - x_0\| \leq \eta$. Take D containing its support and a test f supported in D with $f(x_0) \geq \|x_0\| - \eta \geq 1 - 2\eta$. Choose a norming functional ϕ for x , and impose in addition $\phi(y) = 0$ on the tail space. This is still a closed finite-codimensional subspace. For y in this subspace with $\|y\| \geq 1$, choose $g \in K_p$ with $|g(y)| \geq \|y\|/2$. Select a four-piece increment q as above and multiply it by the sign that points inward from its attachment class. Then $af + \lambda q$ is admissible: its head and tail quadratic budgets are a^2 and λ^2 , and its tail potential changes the inherited value inward by at most 2λ . This test evaluates x at least at $a - 3\eta$, since it has norm at most one and agrees with af on x_0 . Hence

$$\max_{\pm} \|x \pm ty\| \geq 1 - \lambda^2 - 3\eta + t\lambda/8 = 1 + 5t^2/2048 \geq 1 + t^2/512.$$

The functional ϕ gives the lower bound one for both endpoints. Their average therefore has gain at least $t^2/1024$.

There is a different useful normalization of the same four-piece selection in the global model. For a unit $x \in X_{K_g}$, put $\alpha = t/64$ and $\eta = \alpha^2/10$; approximate x by a vector x_0 supported in D within $\eta/10$. A head norming test for x_0 has $f(x) \geq 1 - \eta$. For $y \in F_D$ with $\|y\| \geq 1$, choose an inward piece q with $|q(y)| \geq 1/8$. Its functional norm is at most two, so $|q(x)| \leq 2\eta/10$. The test

$$H = \frac{f + \alpha q}{\sqrt{1 + \alpha^2}}$$

is admissible by disjoint global square sums and the inward path bound. Both endpoint norms are at least $1 - \eta$, and

$$\max_{\pm} \|x \pm ty\| \geq \frac{1 - \eta - 2\alpha\eta/10 + \alpha t/8}{\sqrt{1 + \alpha^2}} \geq 1 + 2\alpha^2. \quad (6)$$

For the last inequality, $\alpha t/8 = 8\alpha^2$ and $\sqrt{1 + \alpha^2} \leq 1 + \alpha^2/2$. The numerator is $1 + (79/10)\alpha^2 - \alpha^3/50$, which dominates $(1 + 2\alpha^2)(1 + \alpha^2/2)$ for $0 \leq \alpha \leq 1/64$. The two endpoint bounds yield average gain at least $19\alpha^2/20 = 19t^2/81920$.

3.3 Independent component selection

For $K = K_a$ every test has finite support. Thus only finitely many exit components carry nonzero increments, and we can choose pieces component by component. We first describe the two choices and their evaluation guarantees, then give their numerical specializations.

For the first choice, take $g \in K_a$ and split its relative potential as

$$R = 2(R_+/2 - (-R)_+/2).$$

The two increment families g^+, g^- have coefficients bounded by $|g_s|/2$ and component potentials in $[0, 1]$, so both belong to K_a . All component sums below are finite. The coefficient identity $g\mathbf{1}_{T \setminus D} = 2(g^+ - g^-)$ shows that one family, denoted k , satisfies $|k(y)| \geq |g(y)|/4$. Let q_j be its evaluation on the j th exit component. Then $\sum_j |q_j| \geq |g(y)|/4$. Choose on each component a sign d_j pointing inward from its inherited head value. For $0 < t < 1$, put $r = t/32$ and $s = \sqrt{1 - r^2}$. Adjoining either rd_jk or zero on each component to sf gives a test. The path value moves inward by at most r , and every antichain receives square sum at most $s^2 + r^2 = 1$. Retain the components with $d_jq_j > 0$ for the plus endpoint and those with $d_jq_j < 0$ for the minus endpoint. For a vector x_0 supported in D , the resulting pair of tests gives

$$\frac{\|x_0 + ty\| + \|x_0 - ty\|}{2} \geq sf(x_0) + \frac{tr}{2} \sum_j |q_j| \geq sf(x_0) + tr|g(y)|/8.$$

For the second choice, split the relative potential of g into its positive and negative parts on each exit component. Select the part whose evaluation has magnitude at least half the magnitude of the full component evaluation, and orient it inward. If the selected increments are h_j , then

$$\sum_j |h_j(y)| \geq |g(y)|/2.$$

Retain either all positively evaluating pieces or all negatively evaluating pieces, according to which group has greater total magnitude. Their sum h satisfies $|h(y)| \geq |g(y)|/4$, $|h_s| \leq |g_s|$, and $h/2 \in K_a$.

All these arrays are finite. For $0 \leq \alpha \leq 1/2$, the single test $\sqrt{1-\alpha^2}f + \alpha h$ is admissible by the same antichain budget and inward path argument.

We now recover the two completed-space estimates. First fix $0 < t < 1$ and a unit x in X_{K_a} . Set $r = t/32$, $s = \sqrt{1-r^2}$, and $\eta = t^2/10000$. Choose a finite x_0 within η of x and a test f supported in a finite initial D with $f(x_0) \geq 1 - 2\eta$. For $y \in F_D$ of norm at least one choose $g \in K_a$ with $g(y) \geq 1/2$. The first choice gives

$$\frac{\|x_0 + ty\| + \|x_0 - ty\|}{2} \geq s(1 - 2\eta) + tr/16.$$

Returning to x loses at most η . Since $s \geq 1 - r^2$, the final excess is at least

$$tr/16 - r^2 - 3\eta = \frac{433}{640000}t^2 > 0. \quad (7)$$

The subspace was fixed before y , as required by the modulus.

For the single-test specialization put $\alpha = t/16$ and $\varepsilon = t^2/2048$. Choose finite x_0 within ε of x and a finite test f with $f(x) > 1 - \varepsilon$; enlarge D to contain both supports. For $y \in F_D$ of norm at least one choose $g \in K_a$ with $g(y) > \|y\|/2$. The second choice gives an inward h with $|h(y)| \geq 1/8$. Since h vanishes on D and has functional norm at most two, $|h(x)| \leq 2\varepsilon$. Evaluating $\sqrt{1-\alpha^2}f + \alpha h$ at a favorable endpoint bounds it below by

$$1 - \alpha^2 - (1 + 2\alpha)\varepsilon + \alpha t/8,$$

while the head test bounds the other by $1 - \varepsilon$. The average excess is consequently at least

$$\frac{\alpha t/8 - \alpha^2 - (2 + 2\alpha)\varepsilon}{2} \geq \frac{5t^2}{4096} \geq \frac{t^2}{1024}.$$

This last choice keeps the full coefficient domination of g and uses only one inward test at the favorable endpoint. The common paired proof and these selection proofs therefore provide different witnesses under the same explicitly stated antichain constraint.

4 A free root and a planar perturbation

We now study an edge norm defined by bounded scalar potentials whose root value is free. The root-inclusive sibling model also permits such a root value, but records it as a root coefficient; here it contributes no coordinate. The rootless model K_s instead fixes the root potential at zero. We give a direct construction of witnesses in the edge coordinates: inward jumps detect the first edges beyond a finite head, and a perturbation in the Euclidean plane detects the remaining edges.

Let $T = \mathbb{N}^{<\omega}$, with root o , and let $T^+ = T \setminus \{o\}$. Identify $s \in T^+$ with the edge from its parent s^- to s , and write $\text{ch}(v)$ for the children of v . Define

$$\mathcal{B} = \left\{ b : T \longrightarrow [-1, 1] : \sum_{w \in \text{ch}(v)} |b_w - b_v|^2 \leq 1 \quad (v \in T) \right\}. \quad (8)$$

For $b \in \mathcal{B}$ and $x \in c_{00}(T^+)$ put

$$f_b(x) = \sum_{s \in T^+} x_s(b_s - b_{s^-}), \quad \|x\|_e = \sup_{b \in \mathcal{B}} f_b(x), \quad (9)$$

and let X_e be the completion. In particular, there is no dummy edge above o and no condition $b_o = 0$.

Every increment in (8) has absolute value at most one. For each $s \in T^+$, the potential equal to one on the subtree rooted at s and zero elsewhere has just one nonzero increment, at s . Together with symmetry of $\mathcal{B}$, these observations give

$$|x_s| \leq \|x\|_e \leq \sum_{u \in T^+} |x_u|, \quad \|e_s\|_e = 1.$$

Thus (9) is a norm and all coordinate functionals are continuous. Each f_b extends to X_e with norm at most one, and the same supremum formula remains valid there: both sides are 1-Lipschitz and agree on the dense space $c_{00}(T^+)$.

Call $D \subset T$ a finite initial subtree if it is finite, contains o , and contains every predecessor of each of its nodes. Given a potential b , retain its labels on D and freeze them at the last node in D along every path leaving D . This preserves admissibility and sets precisely the increments outside D to zero. Consequently the coordinate projection

$$P_D x = \sum_{s \in D \cap T^+} x_s e_s$$

is contractive, and $Q_D = I - P_D$ has norm at most two. The space

$$F_D = \ker P_D = \overline{\text{span}}\{e_s : s \notin D\} \quad (10)$$

is closed and has finite codimension. For the second equality, apply Q_D to finite approximants of a vector in $\ker P_D$. Finally, a finite vector supported in D has a norming potential frozen outside D . Indeed the possible labels on D form a compact set: they lie in $[-1, 1]^D$ and satisfy the finitely many closed child constraints within D ; freezing gives an admissible extension of every such choice.

Theorem 4.1. *The real Banach space X_e defined by (8) and (9) satisfies*

$$\widehat{\delta}_{\|\cdot\|_e}(t) \geq \frac{\sqrt{1+t^2/576} - 1}{4} \quad (0 < t < 1). \quad (11)$$

It admits no equivalent AUC norm. More precisely, if $\alpha \|\cdot\|_e \leq N \leq \beta \|\cdot\|_e$, where $0 < \alpha \leq \beta < \infty$, then $\bar{\delta}_N(\alpha/(4\beta)) = 0$.

The midpoint estimate is proved first for finite centers and tails. The head norming potential already gives a lower bound for both endpoints. We therefore need only increase one endpoint by an amount depending on t . The first-exit and deeper-edge cases supply this increase by different extensions of that potential.

Proof. Fix $0 < t < 1$. Let x be a finite unit vector supported in an initial subtree D , and choose a norming potential b^0 frozen outside D . For every $y \in F_D$,

$$f_{b^0}(x) = 1, \quad f_{b^0}(y) = 0, \quad \|x + ty\|_e, \|x - ty\|_e \geq 1. \quad (12)$$

Suppose first that $y \in F_D$ is finite and $\|y\|_e = 1$. Let $E = \{r \notin D : r^- \in D\}$ be the first-exit nodes. Split $y = y_E + y_L$ into its coordinates on E and on the deeper edges.

First-exit case. Suppose some $c \in \mathcal{B}$ satisfies $|f_c(y_E)| \geq 1/2$. For each $v \in D$ set

$$L_v = \left(\sum_{r \in \text{ch}(v) \setminus D} |y_r|^2 \right)^{1/2}.$$

Cauchy–Schwarz in each sibling group gives $\sum_{v \in D} L_v \geq 1/2$. Choose the inward sign $\sigma_v = -1$ when $b_v^0 \geq 0$ and $\sigma_v = 1$ when $b_v^0 < 0$. For $\varepsilon \in \{-1, 1\}$ define

$$L_{v,\varepsilon} = \left(\sum_{\substack{r \in \text{ch}(v) \setminus D \\ \varepsilon \sigma_v y_r > 0}} |y_r|^2 \right)^{1/2}.$$

Since $L_{v,1} + L_{v,-1} \geq L_v$, one common sign ε has $\sum_v L_{v,\varepsilon} \geq 1/4$.

For $0 < h < 1$, multiply all head labels by $1 - h^2$. At a selected exit r from v , assign the jump

$$d_r = \frac{\varepsilon h y_r}{L_{v,\varepsilon}};$$

assign zero at the other exits, omit groups with $L_{v,\varepsilon} = 0$, and freeze each exit label throughout its descendant subtree. Every nonzero jump has sign σ_v and magnitude at most h . It points toward zero from the head label; if it crosses zero, its resulting magnitude is at most $h < 1$. Thus all labels remain in $[-1, 1]$. At a parent in D the internal and exit increments have total square sum at most

$$(1 - h^2)^2 + h^2 \leq 1,$$

and outside D all further increments vanish. This is an admissible potential. Its evaluations on x and y are respectively $1 - h^2$ and $\varepsilon h \sum_v L_{v,\varepsilon}$. Hence

$$\max\{\|x + ty\|_e, \|x - ty\|_e\} \geq 1 - h^2 + th/4. \quad (13)$$

Taking $h = t/8$ gives the lower bound $1 + t^2/64$.

Deeper-edge case. If the preceding case fails, every admissible test has $|f_c(y_E)| < 1/2$. Choose $c \in \mathcal{B}$ with $f_c(y) > 5/6$; then $f_c(y_L) > 1/3$. We shall preserve the norming head on the horizontal axis and use the vertical coordinate to detect this deeper contribution.

Consider labels $B_v \in \mathbb{R}^2$ satisfying

$$|B_v|_2 \leq 1, \quad \sum_{w \in \text{ch}(v)} |B_w - B_v|_2^2 \leq 1. \quad (14)$$

Every scalar projection of B onto a unit vector belongs to $\mathcal{B}$. Therefore the linear map

$$A_B z = \sum_{s \in T^+} z_s (B_s - B_{s-})$$

satisfies $|A_B z|_2 \leq \|z\|_e$ for finite z , and extends contractively to X_e .

Set $B_v = (b_v^0, 0)$ on D . For each first-exit node r , put $u = b_{r-}^0$ and, throughout the subtree rooted at r , define

$$k = \frac{1}{8}, \quad z_v = k(c_v - c_r), \quad B_v = (u\sqrt{1 - z_v^2}, z_v). \quad (15)$$

The first-exit increment is zero because $z_r = 0$. Also $|z_v| \leq 2k = 1/4$, and $|B_v|_2^2 = u^2(1 - z_v^2) + z_v^2 \leq 1$. For fixed u and c_r , the map

$$H(a) = (u\sqrt{1 - k^2(a - c_r)^2}, k(a - c_r)), \quad -1 \leq a \leq 1,$$

has

$$|H'(a)|_2^2 = k^2 + \frac{u^2 k^4 (a - c_r)^2}{1 - k^2 (a - c_r)^2} \leq \frac{k^2}{1 - 4k^2} = \frac{1}{60}. \quad (16)$$

It is consequently 1-Lipschitz. At a parent outside D this same map applies to all its children, so the child-increment constraint follows from that for c . At a parent in D , the original head increments remain admissible and all first-exit increments are zero. This verifies (14) at the boundary as well as in every descendant subtree.

Now $A_B x = (1, 0)$, whereas the second coordinate of $A_B y$ is $k f_c(y_L) > 1/24$. The first exits contribute zero to that coordinate. The parallelogram identity gives

$$\max\{\|x + ty\|_e^2, \|x - ty\|_e^2\} \geq \frac{|A_B x + t A_B y|_2^2 + |A_B x - t A_B y|_2^2}{2} \quad (17)$$

$$\geq 1 + t^2/576. \quad (18)$$

Completion and arbitrary centers. Put $G(t) = \sqrt{1 + t^2/576} - 1$. Since $G(t) \leq t^2/1152 < t^2/64$, both cases give a maximum endpoint norm at least $1 + G(t)$. Combining this with (12) yields an average at least $1 + G(t)/2$. Finite unit vectors are dense in the unit sphere of F_D , by (10) followed by normalization, so continuity gives this same estimate for every unit $y \in F_D$. For a unit $u \in F_D$, the function

$$r \mapsto \frac{\|x + ru\|_e + \|x - ru\|_e}{2}$$

is convex and even, hence nondecreasing on $[0, \infty)$. The estimate therefore also holds for every $y \in F_D$ with $\|y\|_e \geq 1$. Given an arbitrary unit center, choose a finite unit center within $G(t)/4$ and use its subspace F_D . The reverse triangle inequality loses at most $G(t)/4$ in the average. Taking the infimum over these tail directions, the supremum over finite-codimensional subspaces, and then the infimum over unit centers proves (11), with the direction convention of (1).

The renorming obstruction. For nonroot s , let p_s be the sum of the edges on the path from o to s . Telescoping gives $f_b(p_s) = b_s - b_o$, and the singleton increment test gives the lower bound in

$$1 \leq \|p_s\|_e \leq 2. \quad (19)$$

For every fixed parent s and finite scalar family (a_n) ,

$$\left\| \sum_{n=1}^m a_n e_{s \frown n} \right\|_e \leq \left(\sum_{n=1}^m |a_n|^2 \right)^{1/2}. \quad (20)$$

This follows directly from the sibling constraint in (8). If $\varphi \in X_e^*$, applying (20) with $a_n = \varphi(e_{s \frown n})$ shows

$$\sum_{n=1}^m |\varphi(e_{s \frown n})|^2 \leq \|\varphi\|^2;$$

thus $e_{s \frown n} \rightarrow 0$ weakly against the whole continuous dual. These increments have norm one, and $p_{s \frown n} = p_s + e_{s \frown n}$. Start at any fixed nonroot v and use the tree $(p_{v \frown s})_{s \in T}$. By (19) and (20), Lemma 2.3 applies with $m = 1$, $M = 2$ and $c = 1$. Its obstruction scale is exactly $\alpha c / (2\beta M) = \alpha / (4\beta)$, proving the final assertion. $\square$

5 Linear costs of root paths

The global constraint on a test has a concrete dual meaning. A vector can be paid for either as a Hilbert vector or as a sum of root paths; the norm is the least sum of these two costs. We first prove this exact identification. We then use its two descriptions in different ways: clipping controls dual tests, whereas removal and prediction control signed path coefficients.

5.1 The norm and its finite heads

For an integer $n \geq 1$, let $T_n = \bigcup_{j=0}^n \mathbb{N}^j$, with root $o = \emptyset$ and prefix order $\preceq$. The root belongs to every root path. On the real Hilbert space $H_n = \ell_2(T_n)$ put

$$b_s = \sum_{r \preceq s} e_r, \quad B_n \mu = \sum_{s \in T_n} \mu_s b_s \quad (\mu \in \ell_1(T_n)).$$

Since $\|b_s\|_2 \leq \sqrt{n+1}$, the series defining $B_n \mu$ converges absolutely in H_n and $\|B_n\| \leq \sqrt{n+1}$. Define

$$L_n(x) = \inf_{x=h+B_n\mu} (\|h\|_2 + \|\mu\|_1), \quad (21)$$

$$p_n(f) = \max \left\{ \|f\|_2, \sup_{s \in T_n} \left| \sum_{r \preceq s} f_r \right| \right\}. \quad (22)$$

The infimum ranges over $h \in H_n$ and $\mu \in \ell_1(T_n)$; attainment is not assumed. Restricting μ to finite support gives the same infimum: truncate an arbitrary μ in ℓ_1 and absorb the vanishing remainder $B_n(\mu - \mu^{(k)})$ into h .

Proposition 5.1 (Exact duality for the linear cost). *The space $X_n = (H_n, L_n)$ is complete and*

$$\frac{\|x\|_2}{\sqrt{n+1}} \leq L_n(x) \leq \|x\|_2, \quad L_n(x) = \sup_{p_n(f) \leq 1} \langle f, x \rangle.$$

Every coordinate satisfies $|x_s| \leq L_n(x)$. Its continuous dual is H_n , under the Hilbert pairing, with norm p_n . In particular L_n is exactly the global test norm on T_n .

Proof. The choice $h = x$, $\mu = 0$ proves the upper bound. Every representation of x satisfies

$$\|x\|_2 \leq \|h\|_2 + \sqrt{n+1} \|\mu\|_1 \leq \sqrt{n+1} (\|h\|_2 + \|\mu\|_1),$$

which proves the lower bound. Scaling and adding representations, chosen within an arbitrary error of their infima, prove homogeneity and the triangle inequality. The lower bound gives definiteness; norm equivalence gives completeness and density of finite arrays.

For $p_n(f) \leq 1$, absolute convergence allows us to write

$$|\langle f, x \rangle| \leq \|h\|_2 + \sum_s |\mu_s| \left| \sum_{r \preceq s} f_r \right| \leq \|h\|_2 + \|\mu\|_1.$$

Thus the displayed supremum is at most $L_n(x)$. Conversely, Hahn–Banach gives, for $x \neq 0$, a functional φ such that $\varphi(x) = L_n(x)$ and $|\varphi(z)| \leq L_n(z)$. Since $L_n(z) \leq \|z\|_2$, it has the form $\varphi(z) = \langle f, z \rangle$ with $\|f\|_2 \leq 1$. The representation $b_s = B_n e_s$ gives $L_n(b_s) \leq 1$, hence $|\sum_{r \preceq s} f_r| = |\varphi(b_s)| \leq 1$. This proves equality, including the trivial case $x = 0$.

For an arbitrary $f \in H_n$, the representation estimate gives dual norm at most $p_n(f)$. The Hilbert unit ball and every $\pm b_s$ belong to the L_n unit ball. Testing on these vectors gives both reverse bounds in (22), and therefore the exact dual norm. $\square$

A set $A \subseteq T_n$ is an *initial set* if it contains every prefix of each of its vertices; the empty set is allowed. Write P_A for coordinate restriction to A . Restricting a root path to A gives either zero or the root path ending at its last vertex in A . Grouping coefficients by this endpoint cannot increase

their ℓ_1 norm. Therefore P_A is contractive on X_n . It is also contractive on its dual: restriction freezes every prefix sum at its last retained vertex and decreases the Hilbert norm.

For a finite initial set A , the unit ball of the induced norm on $\mathbb{R}^A$ is

$$\text{conv} \left(B_{\ell_2(A)} \cup \{b_s, -b_s : s \in A\} \right). \quad (23)$$

Indeed this is a compact convex symmetric set whose polar is precisely the set of tests in (22) supported on A . Finite-dimensional separation and Proposition 5.1 identify the balls. The assertion of compactness concerns the finite head only. On the whole block the ball is the norm-closed convex hull of the Hilbert ball and the signed paths: one inclusion follows from convexity, and an almost optimal representation of a vector in the unit ball, followed by finite truncation and a scaling tending to one, proves the other.

At height one, this closed convex hull is exactly the unit ball used by Dilworth et al.: the paths are the root singleton e_o , already in the Hilbert ball, and the vectors $e_o + e_s$ for children s . Under the identification of o with their first coordinate, p_1 is also their dual formula [5, Lemma 2.5]. The higher blocks retain this path geometry through arbitrarily many generations.

We use the outer Hilbert sum

$$X = \left(\bigoplus_{n \geq 1} X_n \right)_{\ell_2}, \quad \|x\|^2 = \sum_n L_n(x_n)^2. \quad (24)$$

A finite initial subset of the disjoint union of the T_n will be called a *finite head*. Its restriction P_A is contractive on X , and such restrictions converge strongly to the identity along an exhaustion. To see the latter assertion, first truncate the block sum, then approximate the remaining coordinates by finite arrays and enlarge their support to an initial set. If $A \subseteq C$ are initial sets, then the window projection

$$R_{C \setminus A} = P_C - P_A \quad \text{satisfies} \quad \|R_{C \setminus A}\| \leq 2.$$

In particular the complementary projection $I - P_A$ has norm at most two; its contractivity is not needed.

Proposition 5.2 (Duality and branching in the outer sum). *The space X is reflexive, has separable dual, and its dual is the Hilbert sum of the spaces (H_n, p_n) , with pairing*

$$f(x) = \sum_n \langle f_n, x_n \rangle, \quad \|f\|_*^2 = \sum_n p_n(f_n)^2.$$

Every coordinate vector $e_{n,s}$ and every root path $b_{n,s}$ has norm one. For fixed n and $|s| < n$, the children $(e_{n,s \frown j})_{j \geq 1}$ are weakly null. If $\alpha \|\cdot\| \leq N \leq \beta \|\cdot\|$, with $0 < \alpha \leq \beta$, then $\bar{\delta}_N(t) = 0$ for every $0 < t \leq \alpha/\beta$. In particular no equivalent norm on X is asymptotically uniformly convex.

Proof. Each block is an equivalently normed Hilbert space and hence reflexive. For completeness of the sum, a Cauchy sequence has a limit in each block; its norm bounds and Cauchy estimates pass to the limit first on finite sets of blocks and then by increasing those sets. Cauchy–Schwarz shows that each displayed dual family defines a bounded functional. Conversely, restrict an arbitrary functional to finitely many blocks, choose almost norming vectors there, and weight them proportionally to the block dual norms. This proves that the sum of their squares is bounded by the square of the original functional norm. Density of finite-block vectors proves the representation and equality of norms. Applying the same argument to the dual sum proves surjectivity of the canonical map into the bidual. Finite arrays are dense in the dual by the same block truncation and Hilbert equivalence, so it is separable.

The cost formula gives $L_n(e_s), L_n(b_s) \leq 1$. The dual vector e_s has $p_n(e_s) = 1$ and evaluates to one on both vectors, giving equality. The weak nullity assertion follows from square summability of every fixed dual array on the n th block.

For clarity we give the renorming argument with its scale and gain. Suppose $\alpha\|\cdot\| \leq N \leq \beta\|\cdot\|$, where $0 < \alpha \leq \beta$, and suppose $d = \bar{\delta}_N(t) > 0$ for some $0 < t \leq \alpha/\beta$. Choose $0 < \eta < d/2$. For each unit center u there is a finite-codimensional closed subspace F such that $N(u + tv) \geq 1 + 2\eta$ on its unit sphere. Convexity on a ray gives $N(u + v) \geq 1 + 2\eta$ whenever $v \in F$ and $N(v) \geq t$.

At an internal vertex s take $u = b_{n,s}/N(b_{n,s})$ and $z_j = e_{n,s \frown j}/N(b_{n,s})$. These vectors are weakly null and $N(z_j) \geq \alpha/\beta \geq t$. Their quotient images in X/F tend to zero, since that quotient is finite-dimensional. Choose $v_j \in F$ with $N(v_j - z_j) \rightarrow 0$. If $N(v_j) < t$, replace v_j by $tv_j/N(v_j)$; for large j this is defined, and the extra change is at most $N(v_j - z_j)$. Thus z_j is approximated by vectors in F of norm at least t . For a sufficiently large child index,

$$N(b_{n,s \frown j}) \geq (1 + \eta)N(b_{n,s}).$$

Iterating to depth n gives $\alpha(1+\eta)^n \leq \beta$, impossible for all n . The asymptotic modulus is nonnegative: the kernel of a norming functional at a unit center has $N(u + tv) \geq 1$. This proves the claimed zero value at every indicated scale. Taking $\eta = d/4$ gives the often useful growth factor $1 + d/4$. $\square$

Two distinctions about the model will matter below. The global test norm on the infinite-height tree restricts to L_n on T_n : restrict an admissible test in one direction, and extend by zero in the other, which freezes all longer prefix sums. This identifies the completed finite-height block, but does not identify the infinite-height completion with (24). Also the quadratic cost

$$Q_n(x) = \inf_{x=h+B_n\mu} (\|h\|_2^2 + \|\mu\|_1^2)^{1/2}$$

satisfies $Q_n \leq L_n \leq \sqrt{2}Q_n$ and is different from L_n : $L_n(e_o) = 1$, whereas $Q_n(e_o) = 1/\sqrt{2}$. The upper bound in the last equality comes from splitting the root coordinate equally; the lower bound follows from $1 \leq \|h\|_2 + \|\mu\|_1 \leq \sqrt{2}(\|h\|_2^2 + \|\mu\|_1^2)^{1/2}$. Thus constants proved for one cost cannot be transferred by identifying the two norms.

5.2 Clipping the tail of a dual test

A head test has a constant prefix sum after leaving the head. We split the coefficients of a second test outside the head into two pieces whose prefix sums point towards zero relative to this constant. Hilbert orthogonality pays for their coefficients, and the inward direction pays for the prefix constraint.

Lemma 5.3 (Inward decomposition for the linear polar). *Let A be a finite head. Suppose $h, g \in X^*$ have norms at most one and $P_A h = h$. There are $r_1, r_2 \in X^*$, vanishing on A , such that*

$$r_1 - r_2 = (I - P_A)g, \quad \|r_l\|_* \leq 2, \quad \|\lambda h + \mu r_l\|_*^2 \leq \lambda^2 + 4\mu^2$$

for $l = 1, 2$ and $\lambda, \mu \geq 0$.

Proof. In block n write $A_n = A \cap T_n$ and set

$$H(s) = \sum_{r \preceq s} h_{n,r}, \quad V(s) = \sum_{\substack{r \preceq s \\ r \notin A_n}} g_{n,r}, \quad a_n = p_n(h_n), \quad c_n = p_n(g_n).$$

Here $|H| \leq a_n$. The sum V is a difference of two prefix sums of g_n , or a single prefix sum if the path misses A_n , so $|V| \leq 2c_n$. It vanishes on A_n . Define

$$(V_1, V_2) = \begin{cases} (\min(V, 0), -\max(V, 0)), & H \geq 0, \\ (\max(V, 0), -\min(V, 0)), & H < 0. \end{cases}$$

Let $r_{l,n,s} = V_l(s) - V_l(s^-)$, with predecessor value zero at the root. These coefficients vanish on A_n . Outside A_n , the head sum H is constant along each remaining branch, and both scalar clipping maps are 1-Lipschitz. Therefore

$$|r_{l,n,s}| \leq |g_{n,s}| \quad (s \notin A_n).$$

At a first exit vertex this follows from the zero predecessor value of V_l . Moreover the prefix sums of $r_{l,n}$ are V_l , and $V_1 - V_2 = V$, proving the claimed difference.

The supports of h_n and $r_{l,n}$ are disjoint. Hence their Hilbert norm satisfies

$$\|\lambda h_n + \mu r_{l,n}\|_2^2 \leq \lambda^2 a_n^2 + \mu^2 c_n^2.$$

The products HV_l are nonpositive, so each prefix sum satisfies

$$|\lambda H + \mu V_l| \leq \max\{\lambda a_n, 2\mu c_n\}.$$

These two estimates give $p_n(\lambda h_n + \mu r_{l,n})^2 \leq \lambda^2 a_n^2 + 4\mu^2 c_n^2$. Summing over n proves the asserted inequality; taking $\lambda = 0$, $\mu = 1$ proves $r_l \in X^*$ and its norm bound. $\square$

Theorem 5.4 (Clipping estimate on a head and its complement). *If $x, y \in X$, $P_A x = x$ and $P_A y = 0$ for a finite head A , then, for every $c > 0$,*

$$\max\{\|x + y\|, \|x - y\|\} \geq \frac{\|x\| + c\|y\|/2}{\sqrt{1 + 4c^2}}. \quad (25)$$

In particular the left side is at least $\sqrt{\|x\|^2 + \|y\|^2/16}$.

Proof. Choose norming dual vectors h, g for x, y , with norms at most one, and replace h by $P_A h$. The projection preserves its evaluation at x and does not increase its norm. Lemma 5.3 gives $r_1(y) - r_2(y) = \|y\|$, so at least one of $|r_1(y)|, |r_2(y)|$ is at least $\|y\|/2$. For this index choose a sign ε with $\varepsilon r_l(y) \geq \|y\|/2$. Because $h(y) = r_l(x) = 0$, evaluation of $h + cr_l$ on $x + \varepsilon y$ proves (25). Its supremum over $c > 0$ is the displayed square root, by Cauchy–Schwarz in two dimensions; endpoint cases follow by a limit. $\square$

The next result has an arbitrary center and no chosen head in its statement. Weak nullity removes the head coordinates of the moving vectors; the approximation of the center must be made before taking the lower limit.

Theorem 5.5 (Clipping along a weakly null sequence). *Let $x \in X$ and let (y_j) be a bounded weakly null sequence in X . If $L = \liminf_j \|y_j\|$, then*

$$\liminf_j \max\{\|x + y_j\|, \|x - y_j\|\} \geq \sqrt{\|x\|^2 + L^2/16}.$$

Proof. Fix $\eta > 0$ and choose a finite head A with $\|x - P_A x\| \leq \eta$, using the strong head exhaustion. Put $z_j = (I - P_A)y_j$. Since P_A has finite rank and y_j is weakly null, $\|P_A y_j\| \rightarrow 0$. In particular,

$$|\|z_j\| - \|y_j\|| \leq \|P_A y_j\| \rightarrow 0.$$

Apply Theorem 5.4 to $P_A x$ and z_j . The triangle inequality then gives

$$\max\{\|x + y_j\|, \|x - y_j\|\} \geq \sqrt{\|P_A x\|^2 + \|z_j\|^2/16} - \eta - \|P_A y_j\|.$$

Keep this head fixed while taking the lower limit in j . The result is at least $\sqrt{\|P_A x\|^2 + L^2/16} - \eta$. Finally let $\eta \downarrow 0$, choosing the heads along the strong exhaustion, so that $\|P_A x\| \rightarrow \|x\|$. This proves the stated bound. $\square$

5.3 Path coefficients on complementary forests

The clipping estimate already controls a displacement supported outside a head. We now examine the path representations in (21) themselves. The deficit between representation cost and the value of a head norming functional on the represented vector bounds the mass whose sign opposes that functional. The remaining endpoint coefficients in each complementary tree have one sign. Positive-mass monotonicity uses that sign directly; the next subsection removes a controlled amount of mass so that the remainder is small in Hilbert norm. These arguments record their own tail bounds and expose coefficient mechanisms that will reappear in the variation and stopping arguments.

Fix one tree T_n , a head A , and $U = T_n \setminus A$. The set U is a forest whose roots are the first vertices outside A . Define its intrinsic path vectors and norm by

$$\begin{aligned} b_t^U &= \sum_{v \in U, v \preceq t} e_v, & B_U \alpha &= \sum_{t \in U} \alpha(t) b_t^U, \\ L_U(z) &= \inf_{z = h + B_U \alpha} (\|h\|_2 + \|\alpha\|_1). \end{aligned}$$

All paths have at most $n + 1$ vertices, so these expressions define a bounded map from $\ell_1(U)$ to $\ell_2(U)$ and a Hilbert-equivalent norm. If v is supported in U , then

$$L_n(v) \leq 2L_U(v|_U). \quad (26)$$

Indeed, after extension by zero, $b_t^U = b_t - P_A b_t$, of cost at most two, while a Hilbert residual extends without changing its cost. Changing sign on any collection of whole trees of U is an isometry for L_U .

Lemma 5.6 (Monotonicity of positive path mass). *If $0 \leq \nu \leq \mu$ in $\ell_1(U)$, then $L_U(B_U \nu) \leq L_U(B_U \mu)$.*

Proof. Choose a norming functional $\phi \in \ell_2(U)$ for $B_U \nu$. Its dual norm is at most one, hence $\|\phi\|_2 \leq 1$ and $F_t := \sum_{v \in U, v \preceq t} \phi(v)$ satisfies $|F_t| \leq 1$. At a root set the value at its nonexistent parent equal to zero, and put

$$\psi(v) = |F_v| - |F_{\text{par}(v)}|.$$

Then $|\psi(v)| \leq |\phi(v)|$, and the prefix sums of ψ are $|F_t|$. Thus ψ also has dual norm at most one: pairing it with a decomposition $h + B_U \alpha$ costs at most $\|h\|_2 + \|\alpha\|_1$. Absolute convergence of the coefficient sums gives

$$L_U(B_U \nu) = \sum_t \nu(t) F_t \leq \sum_t \mu(t) |F_t| = \langle B_U \mu, \psi \rangle \leq L_U(B_U \mu).$$

The claim concerns positive path coefficients; it asserts no general coordinatewise monotonicity of the norm. $\square$

Theorem 5.7 (A tail estimate from positive mass). *Let $C = 6(\sqrt{2} + 1)$. In one component, if $P_A x = x$, $P_A y = 0$, and $L_n(x \pm y) \leq 1$, then*

$$L_n(y) \leq C\sqrt{1 - L_n(x)}. \quad (27)$$

For arbitrary x, y satisfying $P_A x = x$ and $P_A y = 0$, writing $\beta = \max\{L_n(x + y), L_n(x - y)\}$ and $r = L_n(x)$, one has

$$L_n(y)^2 \leq C^2 \beta(\beta - r) \leq C^2(\beta^2 - r^2). \quad (28)$$

Consequently, for a finite head projection P on X ,

$$\|y\|^2 \leq C^2(\|x + y\|^2 + \|x - y\|^2 - 2\|x\|^2) \quad (Px = x, Py = 0). \quad (29)$$

Proof. For the unit estimate set $r = L_n(x)$ and choose a norming functional $g = gP_A$, of norm at most one, with $g(x) = r$; use zero if $x = 0$. For each sign choose $x \pm y = h_{\pm} + B_n \alpha_{\pm}$ of cost at most $1 + \eta$, where $\eta > 0$. Put $d = 1 + \eta - r$. Subtracting the value of g from the cost gives nonnegative terms with sum at most d :

$$\|h_{\pm}\|_2 - g(h_{\pm}) + \sum_t (|\alpha_{\pm}(t)| - \alpha_{\pm}(t)g(b_t)) \leq d.$$

In particular,

$$\|h_{\pm}|_U\|_2 \leq \sqrt{2(1 + \eta)d}.$$

For this bound, use $|g(h_{\pm})| \leq \|P_A h_{\pm}\|_2$ and factor the difference of squares. Its other factor is at most $2(1 + \eta)$. The coefficients satisfying $\alpha_{\pm}(t)g(b_t) \leq 0$ have total mass at most d .

Retain only the other coefficients ending in U , and call their intrinsic forest contributions $v_{\pm}$. Then

$$y|_U = v_+ + w_+, \quad -y|_U = v_- + w_-, \quad L_U(w_{\pm}) \leq K := \sqrt{2(1 + \eta)d} + d.$$

Within each tree of U , the value $g(b_t)$ is constant, since the head of every such path is the same. If that value is zero, no coefficient is retained there. Otherwise both retained arrays have its sign. One isometric change of signs makes both arrays nonnegative. Lemma 5.6 now gives

$$L_U(v_+) \leq L_U(v_+ + v_-) = L_U(w_+ + w_-) \leq 2K.$$

It follows that $L_U(y|_U) \leq 3K$ and $L_n(y) \leq 6K$. Letting $\eta \downarrow 0$, and using $0 \leq 1 - r \leq \sqrt{1 - r}$, proves (27). Scaling gives the first bound in (28) when $\beta > 0$; when $\beta = 0$, both vectors are zero. Contractivity of P_A gives $r \leq \beta$ and the second bound. In each component both endpoint norms are at least the center norm. Thus the maximum of their squared differences from r^2 is at most their sum. Summing over components proves (29), including components with empty heads. $\square$

5.4 Removing mass by subtree quotas

There is another way to use the consistent signs outside a head. Instead of changing a norming functional, remove a small portion of the path coefficients. The remaining path sum can then be bounded in Hilbert norm. The following finite selection fact is the common combinatorial step.

Lemma 5.8 (Subtree quotas). *On a forest of finite height, let μ, ν be nonnegative finitely supported endpoint masses. For a vertex v , let K_v be its descendant set, and put $G(v) = \mu(K_v)$, $W(v) = \nu(K_v)$. There is $0 \leq \rho \leq \mu$ such that*

$$\rho(K_v) \geq \min\{G(v), W(v)\} \quad \text{for every } v, \quad \|\rho\|_1 \leq \|\nu\|_1.$$

The last inequality also holds separately in each tree of the forest.

Proof. Only the finite ancestor hull of the supports matters. Write $q(v) = \min\{G(v), W(v)\}$. Nonnegativity gives

$$q(v) \leq G(v), \quad \sum_{w \text{ child of } v} q(w) \leq q(v).$$

Process vertices from the leaves towards the roots. At v , the mass already marked in its child subtrees totals $\sum_w q(w)$. Mark enough additional, previously unmarked mass of μ below v to make the marked total there equal to $q(v)$; fractions of an endpoint mass are allowed. The two displayed inequalities make this possible. Later steps can only increase a previously met quota. The total finally marked in a tree equals its root quota and is at most its total ν -mass. Take ρ to be the marked mass. $\square$

Theorem 5.9 (The bottom-up tail estimate). *Suppose $0 \leq \varepsilon \leq 1$, $P_A x = x$, $P_A y = 0$, $L_n(x \pm y) \leq 1$, and $L_n(x) \geq 1 - \varepsilon$. Then*

$$L_n(y) \leq 3\sqrt{2\varepsilon} + 8\varepsilon. \quad (30)$$

Moreover the following uniform implication holds on X : for any sequence of finite head projections P_j and vectors satisfying

$$P_j x_j = x_j, \quad P_j y_j = 0, \quad \|x_j \pm y_j\| \leq 1, \quad \|x_j\| \rightarrow 1,$$

one has $\|y_j\| \rightarrow 0$.

Proof. Choose $g = gP_A$ norming x , and finite path representations of both endpoints with cost at most $1 + \eta$. Put $d = 1 + \eta - L_n(x) \leq \varepsilon + \eta$. As in the preceding proof, their Hilbert tails have lengths $K_{\pm} \leq \sqrt{2(1 + \eta)d}$, and their coefficient masses $B_{\pm}$ with $\alpha_{\pm}(t)g(b_t) \leq 0$ satisfy $B_{\pm} \leq d$. Call these coefficients bad and the others good.

In a complementary tree where the common value $c = g(b_t)$ is zero, there are no good coefficients. Otherwise put $\varsigma = \text{sign}(c)$. All good coefficients have sign ς , and all bad nonzero coefficients have sign $-\varsigma$. Apply Lemma 5.8 to the combined good mass μ and combined bad mass ν in each tree, and mark the selected portions with their originating endpoint representation. Denote the selected good mass by ρ , and put $G(v) = \mu(K_v)$, $W(v) = \nu(K_v)$. At a vertex in a tree with $c \neq 0$, the vanishing sum of the two endpoint tails reads

$$h_+(v) + h_-(v) + \varsigma(G(v) - W(v)) = 0.$$

Consequently, in every complementary tree, including those with no good mass,

$$(G(v) - W(v))_+ \leq |h_+(v) + h_-(v)| \quad (v \notin A).$$

After removal, the good tail in either one of the representations has absolute coordinate at most $G(v) - \rho(K_v) \leq (G(v) - W(v))_+$. Its Hilbert norm is therefore at most $K_+ + K_-$. The total removed mass is at most $B_+ + B_-$, and each complementary path costs at most two in L_n . Decomposing the plus endpoint tail into its Hilbert, bad, removed-good, and remaining-good parts gives

$$L_n(y) \leq 2K_+ + K_- + 2B_+ + 2(B_+ + B_-) \leq 3\sqrt{2(1 + \eta)d} + 8d.$$

Let $\eta \downarrow 0$. This proves (30), including its endpoint cases.

We give the outer-sum passage explicitly. For the j -th pair set

$$\begin{aligned} a_{j,n}^{\pm} &= L_n(x_{j,n} \pm y_{j,n}), & r_{j,n} &= L_n(x_{j,n}), \\ c_{j,n} &= \frac{1}{2}(a_{j,n}^+ + a_{j,n}^-), & a_{j,n} &= \max(a_{j,n}^+, a_{j,n}^-). \end{aligned}$$

Convexity gives $c_{j,n} \geq r_{j,n}$, and the endpoint assumptions give $\|a_j^\pm\|_2 \leq 1$. Hence $\|c_j\|_2 \rightarrow 1$. The Hilbert parallelogram identity then gives

$$\|a_j^+ - a_j^-\|_2^2 = 2\|a_j^+\|_2^2 + 2\|a_j^-\|_2^2 - 4\|c_j\|_2^2 \leq 4(1 - \|x_j\|^2) \rightarrow 0.$$

Since $a_j = c_j + |a_j^+ - a_j^-|/2$, we also have $\|a_j\|_2 \rightarrow 1$. Ignore coordinates with $a_{j,n} = 0$, whose vectors are zero, and put $\delta_{j,n} = 1 - r_{j,n}/a_{j,n} \in [0, 1]$. Then

$$\sum_n a_{j,n}^2 \delta_{j,n} \leq \sum_n (a_{j,n}^2 - r_{j,n}^2) \rightarrow 0.$$

Applying the component estimate after division by $a_{j,n}$ gives $L_n(y_{j,n})/a_{j,n} \leq 3\sqrt{2\delta_{j,n}} + 8\delta_{j,n}$; the same ratio is at most one by the triangle inequality. For fixed $\tau > 0$, the squared outer norm is therefore at most

$$(3\sqrt{2\tau} + 8\tau)^2 \sum_n a_{j,n}^2 + \tau^{-1} \sum_n a_{j,n}^2 \delta_{j,n}.$$

First let $j \rightarrow \infty$, then $\tau \downarrow 0$. □

The preceding argument emphasizes uniform decay. A different threshold in the same selection procedure gives a convenient homogeneous squared estimate, without an outer limiting argument.

Theorem 5.10 (A squared estimate by quota removal). *For a finite head projection P on X ,*

$$\|y\|^2 \leq 242(\|x + y\|^2 + \|x - y\|^2 - 2\|x\|^2) \quad (Px = x, Py = 0). \quad (31)$$

The same assertion, with L_n in place of $\|\cdot\|$, holds in each component.

Proof. Work first in one component. Set $r = L_n(x)$ and $M_\pm = L_n(x \pm y)$, so $M_\pm \geq r$. Use a supported norming functional g and finite representations of costs at most $M_\pm + \eta$. Put $d_\pm = M_\pm + \eta - r$. Here a path coefficient is good when $\text{sign}(\alpha_\pm(t))g(b_t) > 1/2$, and bad otherwise. The deficit calculation gives

$$B_\pm \leq 2d_\pm, \quad K_\pm \leq \sqrt{2(M_\pm + \eta)d_\pm}. \quad (32)$$

Write $c = g(b_t)$ for the common value on one complementary tree. If $|c| \leq 1/2$, that tree has no good coefficients. Otherwise put $\varsigma = \text{sign}(c)$; good coefficients have sign ς , and bad nonzero coefficients have sign $-\varsigma$. Let $G_+(v), G_-(v)$ be the subtree masses of the good coefficients from the two representations, and let $W(v)$ be the subtree mass of all bad coefficients. In a tree with $|c| > 1/2$, the endpoint tails satisfy

$$h_+(v) + h_-(v) + \varsigma(G_+(v) + G_-(v) - W(v)) = 0.$$

For the quota selection take $G = G_+$. Since $G_- \geq 0$, the identity gives the following bound, which is also immediate in a tree with no good mass:

$$(G(v) - W(v))_+ \leq |h_+(v) + h_-(v)|.$$

Lemma 5.8 removes at most $B_+ + B_-$ from the plus good coefficients. The remaining plus good tail has Hilbert norm at most $K_+ + K_-$; the removed tail has L_n -norm at most $2(B_+ + B_-)$. Adding the plus Hilbert and bad tails yields

$$\begin{aligned} L_n(y) &\leq 2(K_+ + K_-) + 4(B_+ + B_-) \\ &\leq 11 \sum_{\sigma \in \{+, -\}} \sqrt{(M_\sigma + \eta)d_\sigma}. \end{aligned}$$

Here $d_\sigma \leq M_\sigma + \eta$ and $2\sqrt{2} + 8 < 11$. Squaring the last sum costs at most a factor two, and $(M_\sigma + \eta)d_\sigma \leq (M_\sigma + \eta)^2 - r^2$. Letting $\eta \downarrow 0$ proves the component estimate with $2 \cdot 11^2 = 242$. Summation gives (31). Empty heads cause no exception; their centers are zero. □

5.5 Stopping at the first excess of wrong-sign mass

For a displacement supported in a larger finite head, its part in the smaller head need not vanish. We now average the endpoint decompositions before choosing signs, and measure the mass disagreeing with this average in each head fibre. Stopping where that mass exceeds one quarter of the total gives an estimate for the new coordinates alone.

Theorem 5.11 (A fresh-layer squared estimate). *Let $A \subseteq A'$ be finite ancestor-closed subsets of the disjoint union of the T_n , and put $E = A' \setminus A$. If $P_A x = x$ and $P_{A'} y = y$, then*

$$\|P_E y\|^2 \leq 400 \left(\frac{\|x + y\|^2 + \|x - y\|^2}{2} - \|x\|^2 \right). \quad (33)$$

There is no condition on $P_A y$.

Proof. Write $x^s = x + sy$, $s \in \{-1, 1\}$, and denote averages over s by a bar. Fix $\lambda > 0$. In each nonzero component choose

$$x_n^s = z_n^s + B_n \alpha_n^s, \quad r_n^s = \|z_n^s\|_2 + M_n^s \leq (1 + \lambda) L_n(x_n^s), \quad M_n^s = \|\alpha_n^s\|_1.$$

Use zero decompositions in zero components. Project to A' : truncate the Hilbert terms and push each path endpoint to its last ancestor in A' , discarding endpoints with no such ancestor. Both costs decrease, so all arrays may be taken supported in A' .

In a fixed T_n , let $\pi_A(t)$ be the last ancestor of t in A , or a dummy symbol $\perp$ if none exists. For each fibre set

$$L_b = \sum_{\pi_A(t)=b} \bar{\alpha}_n(t), \quad l_b \in \{-1, 1\}, \quad l_b L_b = |L_b|.$$

Either choice is allowed at zero. Introduce the mass on pairs (s, t) given by $m_n(s, t) = |\alpha_n^s(t)|/2$, and let $h(s, t)$ be the coefficient's sign, chosen arbitrarily at zero. Its wrong-sign mass is

$$W_n = \sum_{s, t: h(s, t) \neq l_{\pi_A(t)}} m_n(s, t).$$

Separating the two signs in each fibre, including the dummy fibre, gives $\bar{M}_n - \sum_b |L_b| = 2W_n$. Since $x_n = P_{A \cap T_n}(\bar{z}_n + B_n \bar{\alpha}_n)$, its component norm is at most $\|P_A z_n^s\|_2 + \sum_b |L_b|$; the dummy term is a harmless nonnegative addition to this cost. It follows that

$$\Delta_n := \bar{r}_n - L_n(x_n) \geq 2W_n + \frac{1}{2} \frac{\overline{\|P_E z_n^s\|_2^2}}{\|z_n^s\|_2} \geq 0, \quad (34)$$

A quotient with zero denominator means zero. The Hilbert estimate used here is

$$\|z\|_2 - \|P_A z\|_2 \geq \frac{\|P_E z\|_2^2}{2\|z\|_2} \quad (z \neq 0, P_{A'} z = z).$$

The quantity $Q = \sum_n \bar{r}_n \Delta_n$ controls both Hilbert leakage and wrong-sign mass. For an array on the disjoint union of the trees, $\|\cdot\|_2$ denotes its Hilbert norm, including summation over components. Since $\|z_n^s\|_2 \leq 2\bar{r}_n$,

$$\begin{aligned} \|P_E \bar{z}\|_2^2 &\leq \overline{\|P_E z^s\|_2^2} \leq 4Q, & \sum_n \Delta_n^2 &\leq Q, \\ Q &\leq \sum_n (\bar{r}_n)^2 - \|x\|^2 \leq (1 + \lambda)^2 \overline{\|x^s\|^2} - \|x\|^2. \end{aligned} \quad (35)$$

The first line follows from (34) and $0 \leq \Delta_n \leq \bar{r}_n$; the second uses $L_n(x_n) \leq \bar{r}_n$ and $(\bar{r}_n)^2 \leq \overline{(r_n^s)^2}$.

It remains to recover the norm of $P_E y$ from this budget. The zero case is immediate. Otherwise take a norming functional for $P_E y$ and precompose it with $P_E = P_{A'} - P_A$. Its coefficient array ϕ , supported in E , satisfies

$$\langle y, \phi \rangle = \|P_E y\|, \quad \|\phi\|_2 \leq 2, \quad \left(\sum_n c_n^2 \right)^{1/2} \leq 2, \quad c_n = \sup_{t \in T_n} \left| \sum_{v \preceq t} \phi(v) \right|. \quad (36)$$

The functional has norm at most two. Testing against Hilbert vectors gives the first coefficient bound. Testing against finite sums of one root path in each component, with arbitrary scalar multipliers, gives the second by the outer Hilbert norm. Approximate each supremum if it is not attained.

The Hilbert part of $\langle y, \phi \rangle = \overline{s \langle x^s, \phi \rangle}$ has absolute value at most $4\sqrt{Q}$, by (35) and (36). For the path part write $g(t) = \sum_{v \preceq t} \phi(v)$ and

$$I_n = \int s h(s, t) g(t) dm_n(s, t).$$

For $v \in E \cap T_n$, let $m(v)$ be the mass of pairs with $t \succeq v$, and $w(v)$ its wrong-sign part. The entire subtree below v is in a single π_A -fibre. Its coefficients therefore satisfy

$$(B_n \bar{\alpha}_n)(v) = l_{\pi_A(v)}(m(v) - 2w(v)) = -\bar{z}(v),$$

where the last equality uses $x(v) = 0$ on E . Thus, whenever $w(v) \leq m(v)/4$,

$$m(v) \leq 2|(B_n \bar{\alpha}_n)(v)| = 2|\bar{z}(v)|.$$

Call a vertex $v \in E$ retained if this inequality's condition holds at v and at every preceding vertex of its path in E . Split $g(t)$ into its sum over retained vertices and the remainder. The remainder starts at the first vertex satisfying $w(v) > m(v)/4$. Such first vertices form an antichain, so their disjoint descendant sets have total mass at most $4W_n$. The remainder is the difference of two prefix sums and has absolute value at most $2c_n$. It therefore contributes at most $8c_n W_n$ to $|I_n|$. The retained contribution is at most $2 \sum_{v \in E \cap T_n} |\phi(v)| |\bar{z}(v)|$. Summing over components and using $2W_n \leq \Delta_n$, we obtain

$$\begin{aligned} \sum_n |I_n| &\leq 4\|P_E \bar{z}\|_2 + 4 \sum_n c_n \Delta_n \\ &\leq 8\sqrt{Q} + 8\sqrt{Q} = 16\sqrt{Q}. \end{aligned}$$

Together with the Hilbert part this proves $\|P_E y\| \leq 20\sqrt{Q}$. Apply (35), square, and let $\lambda \downarrow 0$ to obtain (33). $\square$

6 Energy on newly exposed coordinates

Throughout this section X is the outer Hilbert sum in (24), with block norm L_n on $H_n = \ell_2(T_n)$, root-path vectors b_u , and path map B_n . Proposition 5.1 identifies L_n with the supremum over the tests

$$\|a\|_2 \leq 1, \quad |\langle b_u, a \rangle| \leq 1 \quad (u \in T_n). \quad (37)$$

All coordinate sets below lie in the disjoint union of the trees. A set is *initial* if it contains the ancestors of each of its vertices. We write $R_A = P_A$ for coordinate restriction to A . Recall that R_A is contractive for initial A , and that

$$L_n(R_{B \setminus A} b_u) \leq 2 \quad (A \subseteq B \text{ initial}). \quad (38)$$

Indeed, a root path meets $B \setminus A$ in an interval, whose indicator is the difference of two root-path indicators, with a missing endpoint interpreted as zero.

The estimates in this section measure how much of a dyadic martingale can appear on successive new layers. Predictability means that the layer is chosen before the sign producing the new martingale value. It is this timing, together with the coherence of path coefficients, that makes the estimates uniform in the tree heights. We first prove a bound for martingales supported on predictable heads. We then give a second proof using variation, and two estimates that allow leakage of the parent vector into the new window. The first proof establishes a signed comparison for its child and parent mixtures, using the same positive-mass idea as above in a direct polar argument. The variation proof applies the subtree-quota lemma to its collapsed coefficients.

6.1 Predicting the coefficient of one random path

Fix an integer $m \geq 1$. On $\Omega = \{-1, 1\}^m$, with uniform probability, let $\mathcal{F}_i$ be the sigma-field generated by the first i signs, and write $\mathbb{E}_i$ for conditional expectation. An X -valued dyadic martingale is a sequence $x_i = \mathbb{E}_i x_m$.

Theorem 6.1 (Predictable-layer energy). *Let $x_0, \dots, x_m$ be an X -valued dyadic martingale. Let P_0 be a fixed finite initial set, and, for $1 \leq i \leq m$, let P_i be a finite initial set measurable with respect to $\mathcal{F}_{i-1}$. Suppose, on every history, that*

$$P_{i-1} \subseteq P_i \quad (1 \leq i \leq m), \quad \text{supp } x_i \subseteq P_i \quad (0 \leq i \leq m), \quad \|x_m\| \leq C.$$

Then, for $\Delta_i = P_i \setminus P_{i-1}$,

$$\sum_{i=1}^m \mathbb{E} \|R_{\Delta_i} x_i\|^2 \leq 495 C^2. \quad (39)$$

Proof. There are only finitely many relevant coordinates, since there are finitely many sign histories and all the sets P_i are finite. We first represent the terminal value by one random atom in each component. The finite-head unit ball is exactly the convex hull in (23). This description provides finite random representations by a Hilbert-ball vector or a signed root path.

For each terminal history and each component, apply this description with $P = P_m \cap T_n$, rescaled by $L_n(x_m^n)$. On a finite extension of the probability space we may therefore choose h_n, c_n, u_n so that

$$\mathbb{E}[h_n + c_n b_{u_n} \mid \mathcal{F}_m] = x_m^n, \quad \sum_n (\|h_n\|_2^2 + |c_n|^2) \leq C^2. \quad (40)$$

For each component the sample is either a Hilbert-ball vector h_n , with $c_n = 0$, or a signed root path with $h_n = 0$ and $|c_n| = L_n(x_m^n)$. Zero components use both coefficients zero. An arbitrary root can be used for u_n when $c_n = 0$. The sigma-fields $\mathcal{F}_i$ still record only the original signs. In particular $x_i = \mathbb{E}_i(h_n + c_n b_{u_n})_n$.

Let v_i^n be the deepest vertex of the path to u_n in P_i , or a symbol ∂ if this intersection is empty; set $b_\partial = 0$. The path trace on the new layer is

$$I_i^n = R_{\Delta_i \cap T_n} b_{u_n} = b_{v_i^n} - b_{v_{i-1}^n}, \quad L_n(I_i^n) \leq 2.$$

The sigma-fields $\mathcal{G}_i^n = \sigma(\mathcal{F}_i, v_i^n)$ increase: from v_i^n and the previous sign history one recovers v_{i-1}^n by truncating its path to P_{i-1} . Set $a_i^n = \mathbb{E}[c_n \mid \mathcal{G}_i^n]$. Orthogonality of scalar martingale increments gives

$$\sum_n \sum_{i=1}^m \mathbb{E}|a_i^n - a_{i-1}^n|^2 \leq \sum_n \mathbb{E}|c_n|^2 \leq C^2. \quad (41)$$

We split the new layer into its Hilbert contribution, its old predicted path coefficient, and the change in that prediction:

$$\begin{aligned} R_{\Delta_i} x_i &= H_i + M_i + W_i, \\ H_i &= \mathbb{E}_i R_{\Delta_i}(h_n)_n, \\ M_i &= \mathbb{E}_i(a_{i-1}^n I_i^n)_n, \\ W_i &= \mathbb{E}_i((a_i^n - a_{i-1}^n) I_i^n)_n. \end{aligned}$$

The identity follows because I_i^n is $\mathcal{G}_i^n$ -measurable, so c_n may be replaced by a_i^n inside its conditional expectation. Conditional Jensen, the disjointness of the Δ_i on each outcome, and (41) show that

$$\sum_i \mathbb{E}\|H_i\|^2 \leq C^2, \quad \sum_i \mathbb{E}\|W_i\|^2 \leq 4C^2. \quad (42)$$

Here we use $L_n \leq \|\cdot\|_2$ for the first inequality. Denote conditional expectations given $\mathcal{F}_{i-1}$ by primes. Predictability of Δ_i , together with the support of x_{i-1} , gives $H'_i + M'_i + W'_i = R_{\Delta_i} x_{i-1} = 0$. Consequently

$$\sum_i \mathbb{E}\|M'_i\|^2 \leq 10C^2. \quad (43)$$

It remains to compare the old-coefficient contribution on a child history with its average on the parent history.

We prove $\|M_i\| \leq 4\|M'_i\|$ pointwise. Fix a component and a history through time $i-1$. For $v \in \Delta_i \cap T_n$, let $r(v)$ be the first vertex of Δ_i on its root path. If $v_i^n = v$, then v_{i-1}^n is the parent of $r(v)$, or ∂ when this first vertex is the root. Thus the value of a_{i-1}^n on these outcomes is a scalar $q_{r(v)}$ depending only on this first vertex. It is the conditional mean on the old trace cell determined by the parent history and $v_{i-1}^n = r(v)^-$, with the root case interpreted as ∂ . On a null old trace cell put $q_{r(v)} = 0$; all its parent and child trace probabilities then vanish. Let $\mu(v)$ and $\tilde{\mu}(v)$ be the probabilities of $v_i^n = v$ conditional on this parent history and on a specified child history, respectively. Since both children have probability $1/2$, $\tilde{\mu}(v) \leq 2\mu(v)$.

Put $I(v) = b_v - b_{r(v)^-}$, interpreting the second term as zero when $r(v)$ is the root. Outcomes with $v_i^n \notin \Delta_i$ have zero path trace. The two conditional mixtures are therefore

$$\begin{aligned} M_i^n &= \sum_{v \in \Delta_i \cap T_n} q_{r(v)} \tilde{\mu}(v) I(v), \\ (M'_i)^n &= \sum_{v \in \Delta_i \cap T_n} q_{r(v)} \mu(v) I(v). \end{aligned}$$

For an admissible test a , put $A_v = \sum_{w \in [r(v), v]} a_w$. We have

$$|\langle M_i^n, a \rangle| \leq 2 \sum_{v \in \Delta_i \cap T_n} |q_{r(v)}| \mu(v) |A_v|.$$

Define a test candidate f , zero off $\Delta_i \cap T_n$, by

$$f_v = \text{sign}(q_{r(v)})(|A_v| - |A_{v-}|),$$

where $A_{v-} = 0$ at the first vertex of each interval and $\text{sign}(0) = 0$. The reverse triangle inequality gives $\|f\|_2 \leq \|a\|_2$. Each root path meets Δ_i in at most one interval. Summing f on that interval telescopes to a signed $|A_v|$, which is at most 2, since A_v is the difference of two root-path sums of a . Hence $f/2$ is admissible. Moreover

$$\langle (M'_i)^n, f \rangle = \sum_{v \in \Delta_i \cap T_n} |q_{r(v)}| \mu(v) |A_v|.$$

Taking the supremum over a proves $L_n(M_i^n) \leq 4L_n((M'_i)^n)$. Summing the component squares proves the required comparison. By (43), the total energy of M_i is at most $160C^2$. Finally, $\|H_i + M_i + W_i\|^2 \leq 3(\|H_i\|^2 + \|M_i\|^2 + \|W_i\|^2)$, giving $3(1 + 160 + 4)C^2 = 495C^2$. $\square$

6.2 Variation and removal of positive mass

A different argument keeps all terminal path coefficients rather than sampling one path. The quantity that telescopes is then the variation of their collapsed coefficients. Its gain controls the amount of positive mass that must be removed to make a child layer small in the Hilbert norm.

For a finite initial $R \subset T_n$, define

$$d_R(z, u) = z(u) - \sum_{\substack{v \text{ child of } u \\ v \in R}} z(v), \quad V_R(z) = \sum_{u \in R} |d_R(z, u)|, \quad V_\emptyset(z) = 0.$$

Finite summation up the tree gives

$$R_R z = \sum_{u \in R} d_R(z, u) b_u, \quad V_R(B_n c) \leq \|c\|_1. \quad (44)$$

For the second inequality, each collapsed coefficient sums the c_v whose deepest ancestor in R is its label; these classes are disjoint.

Lemma 6.2 (Removing excess mass). *Let $S \subseteq T \subset T_n$ be finite initial sets, let $z^+, z^- \in H_n$, and put $z = (z^+ + z^-)/2$. Put $U = T \setminus S$ and*

$$g = \frac{1}{2}(V_T(z^+) + V_T(z^-)) - V_S(z).$$

Then $g \geq 0$ and, for either sign,

$$L_n(R_U z^\pm) \leq 2\|R_U z\|_2 + 4g. \quad (45)$$

Proof. If $S = \emptyset$, then $L_n(R_T z^\pm) \leq V_T(z^\pm) \leq 2g$, so the conclusion follows. Otherwise S contains the root. For $u \in T$, write $\pi(u)$ for its deepest ancestor in S . Choose $\sigma_s \in \{-1, 1\}$ so that $\sigma_s d_S(z, s) = |d_S(z, s)|$. Write

$$d_T(z^\pm, u) = \sigma_{\pi(u)}(p_u^\pm - q_u^\pm), \quad p_u^\pm, q_u^\pm \geq 0,$$

using the positive and negative parts. Collapsing the refined path coefficients to S shows that

$$g = 2 \sum_{u \in T} \frac{1}{2}(q_u^+ + q_u^-) \geq 0. \quad (46)$$

For $u \in U$, all descendants in T have the same π -label. Let $P_u^\pm, Q_u^\pm$ be the respective sums of $p_v^\pm, q_v^\pm$ over these descendants. Since $z^\pm(u) = \sigma_{\pi(u)}(P_u^\pm - Q_u^\pm)$,

$$P_u^\pm \leq 2 \text{av } P_u^\pm \leq 2|z(u)| + 2 \text{av } Q_u^\pm, \quad (47)$$

where av means the average of the two signs.

Fix one sign and apply Lemma 5.8 on the finite forest U to the endpoint masses $\mu_u = p_u^\pm$ and $\nu_u = q_u^+ + q_u^- = 2 \text{av } q_u^\pm$. Initiality of S ensures that all descendants in T of a vertex in U remain in U . The corresponding subtree masses are therefore $P_u^\pm$ and $2 \text{av } Q_u^\pm$. The lemma gives $0 \leq r_u \leq p_u^\pm$ with

$$\sum_{v \succeq u} r_v \geq \min\{P_u^\pm, 2 \text{av } Q_u^\pm\}, \quad \sum_{u \in U} r_u \leq 2 \sum_{u \in U} \text{av } q_u^\pm \leq g.$$

By (47), the remaining mass satisfies

$$P_u^\pm - \sum_{v \succeq u} r_v \leq (P_u^\pm - 2 \text{av } Q_u^\pm)_+ \leq 2|z(u)|.$$

Also $\sum_{u \in U} q_u^\pm \leq g$, by (46). The remaining positive mass defines

$$v(u) = \sigma_{\pi(u)} \sum_{w \succeq u} (p_w^\pm - r_w) \quad (u \in U), \quad v = 0 \text{ off } U.$$

Its coordinates satisfy $|v(u)| \leq 2|z(u)|$. The exact decomposition

$$R_U z^\pm = v + \sum_{u \in U} \sigma_{\pi(u)} (r_u - q_u^\pm) R_U b_u$$

therefore costs at most $2\|R_U z\|_2 + 2(g + g)$, as required. $\square$

Theorem 6.3 (Variation energy). *Under all hypotheses of Theorem 6.1,*

$$\sum_{i=1}^m \mathbb{E} \|R_{\Delta_i} x_i\|^2 \leq 252C^2. \quad (48)$$

In particular the bound is 1008 when $\|x_m\| \leq 2$.

Proof. For each terminal history choose, componentwise,

$$x_m^n = h_n + B_n c_n, \quad \|h_n\|_2 + \|c_n\|_1 \leq 2L_n(x_m^n),$$

with both terms zero in a zero component. Only finitely many components are needed. Put $h_i = \mathbb{E}_i(h_n)_n$, $b_i = (B_n \mathbb{E}_i c_n)_n$; then $x_i = h_i + b_i$. The terminal Hilbert and coefficient budgets are each at most $4C^2$:

$$\mathbb{E} \|(h_n)_n\|_{\ell_2(\coprod T_n)}^2 \leq 4C^2, \quad \mathbb{E} \sum_n \|c_n\|_1^2 \leq 4C^2.$$

At time $i-1$, apply Lemma 6.2 in each component to $S = P_{i-1} \cap T_n$, $T = P_i \cap T_n$, and the two values of b_i^n . These sets are fixed before the sign at time i . If $g_{i,n}$ is the variation gain in the lemma, then

$$V_{P_{i-1} \cap T_n}(b_{i-1}^n)^2 + g_{i,n}^2 \leq \mathbb{E}_{i-1} V_{P_i \cap T_n}(b_i^n)^2.$$

Indeed, the average variation equals the previous variation plus the nonnegative number $g_{i,n}$; square and apply scalar Jensen. Telescoping and (44) give

$$\sum_{i,n} \mathbb{E} g_{i,n}^2 \leq 4C^2.$$

Since $R_{\Delta_i} x_{i-1} = 0$, the lemma and the outer Euclidean norm imply

$$\|R_{\Delta_i} x_i\| \leq \|R_{\Delta_i} h_i\|_2 + 2\|R_{\Delta_i} h_{i-1}\|_2 + 4 \left(\sum_n g_{i,n}^2 \right)^{1/2}.$$

For $j = i-1$ and $j = i$, predictability gives $R_{\Delta_i} h_j = \mathbb{E}_j R_{\Delta_i}(h_n)_n$. Conditional Jensen followed by pathwise disjointness of the layers bounds each of the two unscaled Hilbert energies by $4C^2$. Squaring the last three-term inequality and summing now gives $3(1 + 4 + 16)4C^2 = 252C^2$. $\square$

6.3 Outcome-dependent path measures and sign prediction

The next estimate permits an error on the parent window. Here both the positive path mass and its sign may depend on the entire outcome. We place them on one measure space of outcome–path pairs. The scalar martingale of conditional signs then pays for changes of sign prediction, while a stopping antichain controls the remaining positive path mass.

Theorem 6.4 (Adaptive-window energy). *There is an absolute constant K with the following property. Let $x_t = \mathbb{E}_t x_m$, $0 \leq t \leq m$, be an X -valued dyadic martingale. For $0 \leq t < m$, let $A_t \subseteq D_t$ be finite initial sets measurable with respect to $\mathcal{F}_t$, and suppose $D_t \subseteq A_{t+1}$ whenever $t+1 < m$. Put $U_t = D_t \setminus A_t$. Then*

$$\sum_{t=0}^{m-1} \mathbb{E} \|R_{U_t} x_{t+1}\|^2 \leq K \left(\mathbb{E} \|x_m\|^2 + \sum_{t=0}^{m-1} \mathbb{E} \|R_{U_t} x_t\|^2 \right). \quad (49)$$

No support condition on x_t is imposed.

Proof. For every terminal outcome and component choose

$$x_m^n = h_n + B_n \lambda_n, \quad \|h_n\|_2 + \|\lambda_n\|_1 \leq 2L_n(x_m^n).$$

Write $\lambda_n = \epsilon_n \mu_n$, where $\mu_n = |\lambda_n|$, $|\epsilon_n| \leq 1$, and $M_n = \mu_n(T_n)$. Thus μ_n is a positive measure on path endpoints, allowed to depend on the terminal outcome. The two budgets satisfy

$$\mathbb{E} \sum_n (\|h_n\|_2^2 + M_n^2) \leq 4\mathbb{E} \|x_m\|^2. \quad (50)$$

On outcome–endpoint pairs put the finite positive measure

$$\nu_n(\{\omega\} \times \{u\}) = 2^{-m} \mu_n(\omega, \{u\}).$$

This need not be a probability measure; conditional means on its positive-mass cells have their usual meaning. On null cells set them to zero. The operators $\mathbb{E}_t$ still use the original uniform law on sign histories, whereas the conditional means of ϵ_n below use ν_n , which need not give equal mass to the two sign children. Let $a_{t,n}^-$ be the conditional mean of ϵ_n given the first t signs and the truncation of the root path to A_t . Let $a_{t,n}^+$ be its conditional mean given the first $t+1$ signs and its truncation to D_t . Empty truncations are recorded by a dummy symbol. The conditioning partitions, ordered as

$$(0, -), (0, +), (1, -), (1, +), \dots, (m-1, -), (m-1, +),$$

refine successively: the known sign history and truncation of a deeper trace to the earlier head recover the preceding data. Define

$$q_{t,n} = \mathbb{E}_{t+1} \int |a_{t,n}^+ - a_{t,n}^-| d\mu_n.$$

We first establish the weighted sign estimate

$$\sum_{t < m} \mathbb{E} q_{t,n}^2 \leq 4\mathbb{E} M_n^2. \quad (51)$$

For clarity fix n temporarily. Conditional Cauchy–Schwarz gives

$$q_t^2 \leq (\mathbb{E}_{t+1} M) \mathbb{E}_{t+1} \int |a_t^+ - a_t^-|^2 d\mu.$$

Under the original sign law the children are fair, so $M \geq 0$ gives $\mathbb{E}_{t+1}M \leq 2\mathbb{E}_tM$. Put

$$R_t = \max_{0 \leq j \leq t} \mathbb{E}_j M, \quad R_{-1} = 0.$$

The conversion from the sign law to the pair measure is

$$\mathbb{E} \left[(\mathbb{E}_t M) \mathbb{E}_{t+1} \int |a_t^+ - a_t^-|^2 d\mu \right] = \int (\mathbb{E}_t M) |a_t^+ - a_t^-|^2 d\nu,$$

because $\mathbb{E}_t M$ is measurable at time $t+1$. Hence the sum of the integrated right sides is at most

$$2 \sum_{t < m} \int R_t |a_t^+ - a_t^-|^2 d\nu.$$

Expand $R_t = \sum_{j \leq t} (R_j - R_{j-1})$. Each nonnegative increment is constant on each cell C of the partition defining a_j^- . Orthogonality of successive conditional-mean increments under ν gives

$$\begin{aligned} \sum_{t=j}^{m-1} \int_C |a_t^+ - a_t^-|^2 d\nu &\leq \int_C |\epsilon - a_j^-|^2 d\nu \\ &= \int_C |\epsilon|^2 d\nu - \nu(C) |a_j^-|^2 \leq \nu(C). \end{aligned}$$

The left side uses only some increments of the refining sequence; omitting the others preserves the bound. Multiply by the cell-constant value of $R_j - R_{j-1}$, then sum over cells and j . The weighted sum $2 \sum_{t < m} \int R_t |a_t^+ - a_t^-|^2 d\nu$ is consequently at most $2 \int R_{m-1} d\nu = 2\mathbb{E}(MR_{m-1})$.

The scalar input is the finite-horizon form of Doob's L_2 maximal inequality [6]. We include its first-crossing proof:

$$\left\| \max_{j \leq m} \mathbb{E}_j M \right\|_2 \leq 2\|M\|_2.$$

Indeed, if $R = \max_{j \leq m} \mathbb{E}_j M$, partitioning the event $\{R \geq u\}$ according to its first crossing gives $u\mathbb{P}(R \geq u) \leq \mathbb{E}[M \mathbf{1}_{\{R \geq u\}}]$. Integrating in u yields $\mathbb{E}R^2 \leq 2\mathbb{E}(MR)$; Cauchy-Schwarz proves the claim. Thus $2\mathbb{E}(MR_{m-1}) \leq 4\mathbb{E}M^2$, proving (51).

We next compare a child window with its parent window, keeping the component n fixed. For a vertex s , write $C_s = \{u : s \preceq u\}$ for the cone of endpoints whose paths pass through s . Define the signed endpoint measures

$$\lambda'_{t+1} = \mathbb{E}_{t+1}(a_t^- \mu_n), \quad \rho_{t+1} = \mathbb{E}_{t+1}((a_t^+ - a_t^-) \mu_n).$$

For $s \in U_t$, the child sign history together with membership in C_s is measurable for the partition defining a_t^+ . Hence

$$\mathbb{E}_{t+1} \int_{C_s} \epsilon_n d\mu_n = \mathbb{E}_{t+1} \int_{C_s} a_t^+ d\mu_n = \lambda'_{t+1}(C_s) + \rho_{t+1}(C_s).$$

Thus their sum represents the path part of x_{t+1}^n on the window U_t . The error measure satisfies $\|\rho_{t+1}\|_1 \leq q_{t,n}$. Let $h_{t,n} = \mathbb{E}_t h_n$, $\lambda'_t = \mathbb{E}_t \lambda'_{t+1}$, and $\rho_t = \mathbb{E}_t \rho_{t+1}$. Averaging gives, at each $s \in U_t$,

$$x_t^n(s) = h_{t,n}(s) + \lambda'_t(C_s) + \rho_t(C_s). \quad (52)$$

Fix a history through time t . On every cone C_s with $s \in U_t$, truncation to A_t is fixed. Thus a_t^- is constant on that cone, including all completions of the fixed sign history. There is no cancellation within the cone in either λ'_t or λ'_{t+1} . With $\phi_s = |\lambda'_t(C_s)|$, fair binary conditioning therefore gives

$$|\lambda'_{t+1}|(C_s) \leq 2\phi_s. \quad (53)$$

Choose another path decomposition, now only for the parent window:

$$R_{U_t}x_t^n = d + B_n\theta, \quad \|d\|_2 + \|\theta\|_1 \leq 2r_{t,n}, \quad r_{t,n} = L_n(R_{U_t}x_t^n).$$

If $\beta = |\theta| + |\rho_t|$, then (52) gives

$$\phi_s \leq |d(s) - h_{t,n}(s)| + \beta(C_s), \quad \|\beta\|_1 \leq \|\theta\|_1 + \mathbb{E}_t q_{t,n}. \quad (54)$$

Stop along each descending line at the first vertex $s \in U_t$ where $\beta(C_s) \geq \phi_s/2$. These first vertices form an antichain, so their endpoint cones are disjoint. At window vertices neither stopped nor below a stop, $\phi_s \leq 2|d(s) - h_{t,n}(s)|$. By (53), the vector formed by the child's path coordinates at these vertices has Hilbert norm at most $4(\|d\|_2 + \|R_{U_t}h_{t,n}\|_2)$. On the remaining part of the window, every contributing endpoint lies in a stopping cone. Its total $|\lambda'_{t+1}|$ -mass is at most $4\|\beta\|_1$, using the stopping inequality, the factor 2 in (53), and disjointness of the cones. Each path contributes there on an interval, of L_n -norm at most 2. This part therefore costs at most $8\|\beta\|_1$. The error measure ρ_{t+1} costs at most $2q_{t,n}$. Combining these estimates gives the explicit component bound

$$\begin{aligned} L_n(R_{U_t}x_{t+1}^n) &\leq \|R_{U_t}h_{t+1,n}\|_2 + 4\|R_{U_t}h_{t,n}\|_2 \\ &\quad + 16r_{t,n} + 8\mathbb{E}_t q_{t,n} + 2q_{t,n}. \end{aligned} \quad (55)$$

Here $4\|d\|_2 + 8\|\theta\|_1 \leq 16r_{t,n}$.

We have reduced the vector estimate to square summation. Conditional Jensen and predictability of U_t give, for $j = t, t+1$,

$$\mathbb{E}\|R_{U_t}h_{j,n}\|_2^2 \leq \mathbb{E}\|R_{U_t}h_n\|_2^2.$$

The windows are disjoint on every outcome, so summing over t, n bounds either Hilbert energy by $\mathbb{E}\sum_n \|h_n\|_2^2$. The same summation for $q_{t,n}$ and $\mathbb{E}_t q_{t,n}$ is bounded by $4\mathbb{E}\sum_n M_n^2$, by (51) and Jensen. Square the five terms in (55), bounding their sum's square by five times the sum of squares, and use (50). This proves (49) with an absolute constant. All sums over components involve nonnegative terms; Tonelli's theorem justifies them even when the terminal vectors have infinitely many nonzero components. $\square$

6.4 Fixed windows and a direct leakage estimate

For fixed heads, a simpler collapse of the terminal signed masses gives an explicit leakage constant. It uses the mass whose sign disagrees with the previous collapsed coefficient, rather than a martingale of predicted signs. Unlike Theorem 6.1, the conclusion pays for the actual parent contribution to each window.

Theorem 6.5 (Energy with leakage). *Let $W_i = \mathbb{E}_i W_m$, $0 \leq i \leq m$, be an X -valued dyadic martingale. Let $A_0 \subseteq \dots \subseteq A_m$ be fixed finite initial sets and put $\Delta_i = A_i \setminus A_{i-1}$. Then*

$$\sum_{i=1}^m \mathbb{E}\|R_{\Delta_i}W_i\|^2 \leq 1024 \left(\mathbb{E}\|W_m\|^2 + \sum_{i=1}^m \mathbb{E}\|R_{\Delta_i}W_{i-1}\|^2 \right). \quad (56)$$

Proof. We first work in one component, suppressing n and restricting each set to its tree. At each terminal outcome choose

$$W_m = H + B_n c, \quad \|H\|_2 + \|c\|_1 \leq 2L_n(W_m),$$

and let $H_i = \mathbb{E}_i H$. For an endpoint s , let $\pi_i(s)$ be its deepest ancestor in A_i , or ∂ if none exists. Set $\pi_i(\partial) = \partial$ and $b_\partial = 0$. Define

$$c_r^i = \mathbb{E}_i \sum_{\pi_i(s)=r} c_s, \quad v_i = \sum_{r \in A_i \cup \{\partial\}} |c_r^i|.$$

The dummy coefficient is kept in this last sum even though its path vector vanishes. This ensures the exact refinement identity

$$\mathbb{E}_{i-1} \sum_{\pi_{i-1}(r)=q} c_r^i = c_q^{i-1} \quad (q \in A_{i-1} \cup \{\partial\}). \quad (57)$$

The child window itself has the representation

$$R_{\Delta_i} W_i = R_{\Delta_i} H_i + \sum_r c_r^i R_{\Delta_i} b_r.$$

Orient each refined coefficient by its old collapsed coefficient:

$$\sigma_r = \text{sign}(c_{\pi_{i-1}(r)}^{i-1}), \quad c_r^i = \sigma_r (g_r - q_r), \quad g_r, q_r \geq 0,$$

where sign at zero is chosen to be $+1$, and g_r, q_r are the positive and negative parts of $\sigma_r c_r^i$. The signs are known on the parent history. By (57),

$$\beta_i := \mathbb{E}_{i-1} \sum_r q_r = \frac{1}{2} (\mathbb{E}_{i-1} v_i - v_{i-1}) \geq 0.$$

The elementary inequality $(\mathbb{E}_{i-1} v_i - v_{i-1})^2 \leq \mathbb{E}_{i-1} v_i^2 - v_{i-1}^2$ therefore telescopes to

$$\sum_i \mathbb{E} \beta_i^2 \leq \frac{1}{4} \mathbb{E} v_m^2 \leq \frac{1}{4} \mathbb{E} \|c\|_1^2. \quad (58)$$

Fix a step and a parent history. For $t \in \Delta_i$, all its descendant labels in A_i have the same old ancestor and therefore the same sign σ_t . Set $g(t) = \sum_{r \succeq t} g_r$ and $q(t) = \sum_{r \succeq t} q_r$, with the dummy omitted. Averaging the coordinate formula gives

$$W_{i-1}(t) = H_{i-1}(t) + \sigma_t (\mathbb{E}_{i-1} g(t) - \mathbb{E}_{i-1} q(t)).$$

Decompose the leaked parent window once more:

$$Y = R_{\Delta_i} W_{i-1} = L + B_n d, \quad \|L\|_2 + \|d\|_1 \leq 2L_n(Y).$$

There are two equally likely children and $g(t) \geq 0$, so

$$g(t) \leq 2\mathbb{E}_{i-1} g(t) \leq 2 \left(|L(t)| + |H_{i-1}(t)| + \sum_{s \succeq t} |d_s| + \mathbb{E}_{i-1} q(t) \right).$$

Call $t \in \Delta_i$ large when $g(t) > 4(|L(t)| + |H_{i-1}(t)|)$. At a large node,

$$g(t) \leq 4 \left(\sum_{s \succeq t} |d_s| + \mathbb{E}_{i-1} q(t) \right).$$

The minimal large nodes form an antichain. Mark every coefficient label in A_i below one of them. Their descendant sets are disjoint, so the total marked positive mass is at most $4(\|d\|_1 + \beta_i)$. The total negative mass satisfies $\sum_r q_r \leq 2\beta_i$, again by fair binary averaging. Using $L_n(R_{\Delta_i} b_r) \leq 2$, all marked positive terms together with all negative terms cost at most $8\|d\|_1 + 12\beta_i$.

It remains to estimate the unmarked positive part. Its coordinate is zero at every large vertex, since all descendant labels there are marked. At other vertices its absolute value is at most $g(t) \leq 4(|L(t)| + |H_{i-1}(t)|)$. Its Hilbert norm is consequently at most $4(\|L\|_2 + \|R_{\Delta_i} H_{i-1}\|_2)$. Including the child Hilbert term yields

$$\begin{aligned} L_n(R_{\Delta_i} W_i) &\leq \|R_{\Delta_i} H_i\|_2 + 4\|R_{\Delta_i} H_{i-1}\|_2 + 12\beta_i \\ &\quad + 16L_n(R_{\Delta_i} W_{i-1}). \end{aligned} \tag{59}$$

The last coefficient follows from $4\|L\|_2 + 8\|d\|_1 \leq 16L_n(Y)$.

For $j = i-1, i$, conditional Jensen and the fixed, disjoint windows give $\sum_i \mathbb{E} \|R_{\Delta_i} H_j\|_2^2 \leq \mathbb{E} \|H\|_2^2$. Squaring the four terms in (59), using four times their squared sum, and applying (58), gives

$$\sum_i \mathbb{E} L_n(R_{\Delta_i} W_i)^2 \leq 68\mathbb{E} \|H\|_2^2 + 144\mathbb{E} \|c\|_1^2 + 1024 \sum_i \mathbb{E} L_n(R_{\Delta_i} W_{i-1})^2.$$

The first two terms are at most $144\mathbb{E}(\|H\|_2 + \|c\|_1)^2 \leq 576\mathbb{E} L_n(W_m)^2$. Finally sum over the components and replace 576 by 1024 in the terminal term. This proves (56). $\square$

7 Quadratic path costs

7.1 The quotient norm and its normalization

For an integer $n \geq 1$, let $T_n = \mathbb{N}^{\leq n}$, including the empty sequence o , with the initial-segment order $\preceq$. Work over the real field. In $H_n = \ell_2(T_n)$ put

$$p_t = \sum_{s \preceq t} e_s, \quad J_n \lambda = \sum_{t \in T_n} \lambda_t p_t \quad (\lambda \in \ell_1(T_n)).$$

The series converges absolutely in H_n , since $\|p_t\|_2 \leq \sqrt{n+1}$. Thus $(J_n \lambda)(s) = \sum_{t \succeq s} \lambda_t$ and $\|J_n\| \leq \sqrt{n+1}$. Define Z_n to be H_n with the norm

$$|z|_n = \inf_{z=h+J_n \lambda} (\|h\|_2^2 + \|\lambda\|_1^2)^{1/2}, \quad h \in H_n, \quad \lambda \in \ell_1(T_n), \tag{60}$$

and set

$$X_q = \left(\bigoplus_{n \geq 1} Z_n \right)_2, \quad \|z\|_q^2 = \sum_n |z_n|_n^2.$$

In the notation of Section 5, $p_t = b_t$, $J_n = B_n$, and $|z|_n = Q_n(z)$. To bound midpoint tails, we compare the costs of two endpoint representations with the cost of their average restricted to a finite head.

Proposition 7.1. *The spaces Z_n and X_q are Banach and reflexive, and finite node support is dense in X_q . For $z \in H_n$,*

$$\frac{\|z\|_2}{\sqrt{n+2}} \leq |z|_n \leq \|z\|_2, \quad |z_t| \leq \sqrt{2} |z|_n. \quad (61)$$

In particular $1/\sqrt{2} \leq |p_t|_n, |e_t|_n \leq 1$, and $|e_o|_n = 1/\sqrt{2}$. For $f \in \ell_2(T_n)$ the dual norm is

$$|f|_{n,*} = \left(\|f\|_2^2 + \sup_{t \in T_n} \left| \sum_{s \preceq t} f_s \right|^2 \right)^{1/2}. \quad (62)$$

Proof. Scaling and adding representations in (60) proves homogeneity and the triangle inequality. Every representation satisfies

$$\|z\|_2 \leq \|h\|_2 + \sqrt{n+1} \|\lambda\|_1 \leq \sqrt{n+2} (\|h\|_2^2 + \|\lambda\|_1^2)^{1/2};$$

the choice $h = z$ gives the upper bound. This proves positive definiteness, completeness, and equivalence to the Hilbert norm. The dual and bidual therefore have the same underlying continuous linear functionals as for H_n , so Z_n is reflexive. For each coordinate, $|z_t| \leq \|h\|_2 + \|\lambda\|_1$, which gives the second assertion of (61). A single atomic mass represents p_t with cost one. For e_t use its Hilbert representation. At the root, the representation $e_o = e_o/2 + J_n(\delta_o/2)$ attains the matching cost $1/\sqrt{2}$.

The quotient map $(h, \lambda) \mapsto h + J_n \lambda$ has domain $H_n \oplus_2 \ell_1(T_n)$. Pulling the functional f back gives $(f, J_n^* f)$, whose norm is $(\|f\|_2^2 + \|J_n^* f\|_\infty^2)^{1/2}$. Indeed the upper bound is Cauchy–Schwarz, and the reverse bound follows by choosing the Hilbert vector in the direction of f , an atomic coordinate arbitrarily close to the supremum for $J_n^* f$, and then optimizing their two nonnegative lengths. The pullback and quotient functional norms agree: a representation bounds the functional by its cost, and representations can have cost arbitrarily close to $|z|_n$. Since $(J_n^* f)(t) = \sum_{s \preceq t} f_s$, this proves (62) without requiring a maximizing endpoint.

For completeness of the outer sum, a Cauchy sequence converges in each Z_n ; its Cauchy bounds pass first to every finite sum and then to the whole sum, proving membership and convergence of the limit. A bounded functional on this sum restricts to functionals $f_n \in Z_n^*$ with $\sum_n \|f_n\|^2 < \infty$. To prove the bound on this sum, test on finite collections of almost norming unit vectors multiplied by arbitrary Euclidean unit scalar vectors. Conversely such (f_n) defines a functional by Cauchy–Schwarz. Density of finite component sums shows that these constructions are inverse and isometric. Applying the same description to the bidual and using component reflexivity proves reflexivity of X_q . Finally truncate the outer sum and then the Hilbert coordinates in each remaining component to obtain finite node support. $\square$

Write

$$H = \left(\bigoplus_n H_n \right)_2, \quad L = \left(\bigoplus_n \ell_1(T_n) \right)_2.$$

The componentwise map $J : L \rightarrow X_q$ and the coordinate inclusion $H \rightarrow X_q$ are contractions, and

$$\|z\|_q = \inf_{z=h+J\lambda} (\|h\|_H^2 + \|\lambda\|_L^2)^{1/2}. \quad (63)$$

For the reverse inequality in this formula choose component representations with total error in the squared costs less than any prescribed positive number. This also produces $h \in H$ and $\lambda \in L$. The map J here takes values in X_q ; no uniform bound $L \rightarrow H$ is used.

A *finite ancestral head* is a finite set A of nodes in the disjoint union of the T_n , closed under taking ancestors within each component. This is the closure condition called initial in the preceding sections. Its intersection A_n with T_n may be empty. Let P_A retain the coordinates in A and let $Q_A = I - P_A$. For $t \in T_n$ denote its longest ancestor in A_n by $r(t)$, using $\perp$ when there is none, and put $p_\perp = 0$.

Lemma 7.2. *The operator R_A which sends the mass λ_t to $r(t)$ and discards it when $r(t) = \perp$ is contractive on L . Moreover*

$$P_A J = J R_A, \quad \|P_A\| \leq 1, \quad \|Q_A\| \leq 2, \quad |Q_A J_n \lambda|_n \leq 2 \|\lambda\|_1.$$

Here the last formula is in a single component. In particular the path $p_t - p_{r(t)}$ outside the head has norm at most two.

Proof. The triangle inequality bounds the absolute sum in every fiber of r , so R_A contracts each ℓ_1 component and hence L . Coordinate restriction gives $P_A p_t = p_{r(t)}$ and thus $P_A J = J R_A$. Projecting h orthogonally in H and folding λ in (63) proves contractivity of P_A . The remaining bounds follow from $Q_A = I - P_A$ and $\|J\lambda\|_q \leq \|\lambda\|_L$. $\square$

Proposition 7.3. *No equivalent norm on X_q is asymptotically uniformly convex. More precisely, if $a\|z\|_q \leq N(z) \leq b\|z\|_q$, with $0 < a \leq b < \infty$, then $\bar{\delta}_N(t_0) = 0$ at $t_0 = a/(\sqrt{2}b)$.*

Proof. For each fixed nonterminal $s \in T_n$, the coordinate vectors $e_{s \frown j}$ are weakly null in Z_n , by its equivalence to H_n , and therefore in X_q . Their norms and the norms of p_s are at least $1/\sqrt{2}$, while $\|p_s\|_q \leq 1$. Suppose $\bar{\delta}_N(t_0) > 0$ and choose $c > 0$ with $2c < \bar{\delta}_N(t_0)$. At $u = p_s/N(p_s)$ choose a closed finite-codimensional subspace F such that $N(u + t_0 v) > 1 + 2c$ for every $v \in F$ with $N(v) = 1$. The normalized sequence $v_j = e_{s \frown j}/N(e_{s \frown j})$ is weakly null, because its denominators are bounded below. Its image in X_q/F is norm null. Approximation in F , followed by normalization, gives vectors of $F \cap S_N$ whose N -distance from v_j tends to zero. Consequently $N(u + t_0 v_j) > 1 + c$ for all sufficiently large j .

For a convex function $r \mapsto N(u + r v_j)$ with value one at zero, a lower bound greater than one at t_0 persists for $r \geq t_0$. Apply this with $r = N(e_{s \frown j})/N(p_s) \geq a/(\sqrt{2}b)$ to obtain $N(p_{s \frown j}) \geq (1 + c)N(p_s)$. Choosing successors for n steps gives $b \geq (1 + c)^n a/\sqrt{2}$, impossible for arbitrarily large n . The asymptotic modulus is nonnegative: for each unit center one may restrict to the kernel of a norming functional. It is therefore zero. $\square$

7.2 What is lost by averaging and restricting

Fix a finite ancestral head A , write $P = P_A$, $Q = Q_A$, and suppose $Px = x$. Choose representations

$$x \pm y = h^\pm + J\lambda^\pm.$$

We will always choose finite total squared cost, as allowed by (63). Put

$$\bar{h} = \frac{1}{2}(h^+ + h^-), \quad \tilde{h} = \frac{1}{2}(h^+ - h^-), \quad \sigma = \frac{1}{2}(\lambda^+ + \lambda^-), \quad \tau = \frac{1}{2}(\lambda^+ - \lambda^-),$$

and, in each component, put

$$M_n = \frac{1}{2}(\|\lambda_n^+\|_1 + \|\lambda_n^-\|_1), \quad \nu_n = (R_A \sigma)_n, \quad q_n = \|\nu_n\|_1, \quad d_n = M_n - q_n.$$

Thus $x = P\bar{h} + J\nu$ and $d_n \geq 0$. To separate the two sources of this loss, view $m_n = (|\lambda_n^+| + |\lambda_n^-|)/2$ as a nonnegative endpoint measure and σ_n as a signed endpoint measure. Let $C_r = \{t : r(t) = r\}$ for $r \in A_n \cup \{\perp\}$. Absolute convergence gives

$$d_n = \sum_{r \in A_n} (m_n(C_r) - |\sigma_n(C_r)|) + m_n(C_\perp).$$

Each term is nonnegative. The first sum records cancellation during averaging and folding within retained fibres; the last term records the mass discarded when the head is empty in that component.

Lemma 7.4. *Let*

$$C = \frac{1}{2} \sum_{\epsilon \in \{+, -\}} (\|h^\epsilon\|_H^2 + \|\lambda^\epsilon\|_L^2), \quad E = C - \|x\|_q^2.$$

Then

$$\|Q\bar{h}\|_H^2 + \|\tilde{h}\|_H^2 + \sum_n (M_n^2 - q_n^2) \leq E, \quad (64)$$

$$\frac{1}{2}(\|Qh^+\|_H^2 + \|Qh^-\|_H^2) + \sum_n d_n^2 \leq E. \quad (65)$$

Proof. The Hilbert parallelogram identity and the scalar square inequality give $C \geq \|\bar{h}\|_H^2 + \|\tilde{h}\|_H^2 + \sum_n M_n^2$. On the other hand the displayed representation of x has squared cost $\|P\bar{h}\|_H^2 + \sum_n q_n^2 \geq \|x\|_q^2$. Subtract, using the orthogonal decomposition in H , to prove (64). Now $M_n^2 - q_n^2 \geq (M_n - q_n)^2$ and

$$\frac{1}{2}(\|Qh^+\|_H^2 + \|Qh^-\|_H^2) = \|Q\bar{h}\|_H^2 + \|Q\tilde{h}\|_H^2.$$

These identities imply (65). $\square$

To convert this loss into a bound for the vector tail, we need an elementary representation of a positive array on the outside forest. Its importance is that the bound is independent of the height and allows countably many children.

Lemma 7.5 (Stopped flows). *Let $O = T_n \setminus A_n$, let $\mathcal{R}$ be the roots of its connected subtrees, and suppose $V : O \rightarrow [0, \infty)$ satisfies*

$$V(t) \geq \sum_{c \text{ child of } t} V(c), \quad B := \sum_{c \in \mathcal{R}} V(c) < \infty.$$

For any signs ε_t constant on each connected subtree of O , the vector $\varepsilon_t V(t)$ on O , extended by zero on A_n , has norm at most $2B$ in Z_n . The pointwise minimum of two arrays satisfying the child inequality also satisfies it.

Proof. Set $\alpha_t = V(t) - \sum_{c \text{ child of } t} V(c) \geq 0$. Induction from the last level, using nonnegative sums, gives $V(t) = \sum_{s \geq t} \alpha_s$ and $\sum_{t \in O} \alpha_t = B$. This induction is over the finite height, not over a finite set of children. Hence the required vector is

$$\sum_{t \in O} \varepsilon_t \alpha_t (p_t - p_{r(t)}).$$

The series is absolutely convergent in Z_n by Lemma 7.2 and has norm at most $2B$. Continuity of the coordinates identifies its sum with the stated array. Finally

$$\sum_c \min(V(c), W(c)) \leq \min\left(\sum_c V(c), \sum_c W(c)\right) \leq \min(V(t), W(t)),$$

which proves the assertion about minima. $\square$

7.3 Clipping one endpoint against the minority mass

Theorem 7.6. *Let A be a finite ancestral head in X_q and let $P_A x = x$. For every $y \in X_q$ and $R \geq 0$,*

$$\|x + y\|_q, \|x - y\|_q \leq R \implies \|Q_{Ay}\|_q \leq 8\sqrt{R^2 - \|x\|_q^2}. \quad (66)$$

The displacement y need not vanish on the head.

Proof. Convexity gives $\|x\|_q \leq R$, so the right side is defined. The case $R = 0$ is immediate. Choose the representations above with each squared cost at most $R^2 + \eta$, where $\eta > 0$. Then $E \leq D := R^2 + \eta - \|x\|_q^2$.

Fix a component. For every fiber of $r(t)$ choose its majority sign ε_r among the coefficients of both labeled arrays λ^+, λ^- . If the positive and negative masses in a nondiscarded fiber are a_r, b_r , its contribution to d_n is $\min(a_r, b_r)$; for a discarded fiber its contribution is $(a_r + b_r)/2 \geq \min(a_r, b_r)$. Thus the total minority mass over the two arrays is at most d_n .

Outside A_n , each descendant has the same $r(t)$ as its ancestor. For $\epsilon \in \{+, -\}$ define the majority and minority occupancies

$$U^\epsilon(t) = \sum_{s \geq t} (\varepsilon_{r(s)} \lambda_n^\epsilon(s))_+, \quad V^\epsilon(t) = \sum_{s \geq t} (-\varepsilon_{r(s)} \lambda_n^\epsilon(s))_+, \quad V = V^+ + V^-.$$

Here $a_+ = \max(a, 0)$. All these arrays satisfy the child inequality of Lemma 7.5, and the total initial mass of V is at most d_n . Since x vanishes outside the head,

$$\varepsilon_{r(t)}(U^+(t) + U^-(t) - V(t)) = -h_n^+(t) - h_n^-(t).$$

Let $S(t) = \min(U^+(t), V(t))$. It is a stopped flow with initial mass at most d_n , and

$$0 \leq U^+(t) - S(t) \leq (U^+(t) + U^-(t) - V(t))_+ \leq |h_n^+(t) + h_n^-(t)|.$$

Consequently the tail of y_n , which agrees with that of $x_n + y_n$, is the sum of the signed flows $S, -V^+$ and the Hilbert vector $h_n^+ + \varepsilon_r(U^+ - S)$, all restricted to the outside. Lemma 7.5 gives

$$\|Q_{Ay_n}\|_n \leq 4d_n + \|Q_A h_n^+\|_2 + \|Q_A(h_n^+ + h_n^-)\|_2.$$

Take the outer Euclidean norm and use (65). The three terms are at most $4\sqrt{D}$, $\sqrt{2D}$, and $2\sqrt{D}$, respectively. Thus $\|Q_{Ay}\|_q \leq (6 + \sqrt{2})\sqrt{D} \leq 8\sqrt{D}$. Let $\eta \downarrow 0$. No minimizing representation was assumed. $\square$

The next corollary quantifies the symmetric-lens geometry in the characterization of AMUC by Dilworth et al. [5, Theorem 2.1].

Corollary 7.7. *If $x \in X_q$ and an infinite sequence (y_j) satisfies $\|x \pm y_j\|_q \leq R$ and $\|y_i - y_j\|_q \geq \varepsilon > 0$ for $i \neq j$, then*

$$\|x\|_q^2 + \frac{\varepsilon^2}{256} \leq R^2.$$

Proof. Approximate x within $\rho > 0$ by x_0 supported in a finite ancestral head A . Theorem 7.6 bounds each Q_{Ay_j} by $8\sqrt{(R + \rho)^2 - \|x_0\|_q^2}$. The sequence (y_j) is bounded, and P_A has finite rank, so two distinct head projections can be arbitrarily close. Their difference and the two tail bounds give $\varepsilon \leq 16\sqrt{(R + \rho)^2 - \|x_0\|_q^2}$. Let $\rho \downarrow 0$. $\square$

The arbitrary-tail estimate controls the complement of a head without requiring the displacement to vanish on the head. The remaining arguments organize the same cancellation budget for other purposes: symmetric clipping keeps the average of the two squared endpoint norms; stopping and majority pruning retain different parts of the signed mass; and a componentwise passage produces a finite-codimensional midpoint estimate.

7.4 A symmetric clipping estimate

The proof clips both endpoint arrays while keeping the two Hilbert variables separate.

Proposition 7.8. *For a finite ancestral head A , $P_A x = x$, and $P_A y = 0$,*

$$\|x\|_q^2 + \frac{\|y\|_q^2}{144} \leq \frac{\|x+y\|_q^2 + \|x-y\|_q^2}{2}. \quad (67)$$

Proof. It suffices to prove the assertion in one component, since the result can then be summed. Choose endpoint representations whose squared costs exceed $|x \pm y|_n^2$ by at most $\eta > 0$, and use the notation of Lemma 7.4, omitting the component index. Put $E = (|x+y|_n^2 + |x-y|_n^2)/2 + \eta - |x|_n^2$. Then

$$\|Q\bar{h}\|_2^2 + \|\tilde{h}\|_2^2 + M^2 - q^2 \leq E.$$

Orient each nondiscarded fiber by the sign of its folded mean coefficient $\nu_{r(t)}$, choosing $+1$ when it is zero, and choose either sign in $\{-1, 1\}$ on a discarded fiber. Define half-weighted positive and negative endpoint masses by

$$\alpha_t^\epsilon = \frac{1}{2}(\varepsilon_{r(t)}\lambda_t^\epsilon)_+, \quad \beta_t^\epsilon = \frac{1}{2}(-\varepsilon_{r(t)}\lambda_t^\epsilon)_+, \quad \epsilon \in \{+, -\}.$$

Let $\beta = \beta^+ + \beta^-$. In a nondiscarded fiber the positive mass minus negative mass is the absolute folded coefficient. The discarded fiber contributes zero to q . It follows that $\|\beta\|_1 \leq M - q \leq \sqrt{E}$.

For $t \notin A_n$ put $A^\epsilon(t) = \sum_{s \geq t} \alpha_s^\epsilon$ and $B^\epsilon(t) = \sum_{s \geq t} \beta_s^\epsilon$, and let $A_{\text{tot}} = A^+ + A^-$, $B_{\text{tot}} = B^+ + B^-$. The endpoint identities give

$$0 = \bar{h}(t) + \varepsilon_{r(t)}(A_{\text{tot}}(t) - B_{\text{tot}}(t)), \quad y(t) = \tilde{h}(t) + \varepsilon_{r(t)}(A^+(t) - A^-(t) - B^+(t) + B^-(t)).$$

For each sign put $S^\epsilon = \min(A^\epsilon, B_{\text{tot}})$ and $V^\epsilon = A^\epsilon - S^\epsilon$. Then $0 \leq V^\epsilon \leq (A_{\text{tot}} - B_{\text{tot}})_+ \leq |\bar{h}|$ outside the head. Each S^ϵ is a stopped flow of initial mass at most $\|\beta\|_1$, and the same bound applies to each B^ϵ . Lemma 7.5 estimates their four signed vectors by $8\|\beta\|_1$ in total. The remaining terms have Hilbert norm at most $\|Q\tilde{h}\|_2 + 2\|Q\bar{h}\|_2$. Since y vanishes on the head,

$$|y|_n \leq \|Q\tilde{h}\|_2 + 2\|Q\bar{h}\|_2 + 8\|\beta\|_1 \leq 12\sqrt{E}.$$

Let $\eta \downarrow 0$, and sum the resulting component inequalities. Empty heads cause no exception: their entire mass is in the discarded fiber, and the same bounds hold. $\square$

Corollary 7.9. *If $\|x \pm y_j\|_q \leq r$ and $\|y_i - y_j\|_q \geq 2d > 0$ for $i \neq j$ in an infinite sequence, then*

$$\|x\|_q^2 + d^2/144 \leq r^2.$$

Proof. Choose x_0 of finite node support with $\|x - x_0\|_q < \epsilon$, and a finite ancestral head A supporting it. Finite rank and boundedness give distinct i, j with $\|P_A z\|_q < \epsilon$ for $z = (y_i - y_j)/2$. Convexity gives $\|x \pm z\|_q \leq r$, while $\|z\|_q \geq d$. Apply (67) to $x_0, Q_A z$. These vectors have endpoint norms at most $r + 2\epsilon$, and $\|Q_A z\|_q \geq d - \epsilon$. Let $\epsilon \downarrow 0$. $\square$

7.5 Stopping where signed mass cancels

Instead of orienting the coefficients, one can stop where their signed sum is less than half of their absolute mass. The discarded subtrees are disjoint, so their total mass is paid for by the same loss d_n . This gives a smaller numerical constant by a different decomposition.

Proposition 7.10. *Under the hypotheses of Theorem 7.6,*

$$\|Q_A y\|_q \leq 7\sqrt{R^2 - \|x\|_q^2}.$$

Proof. Choose the endpoint representations with squared costs at most $(R + \eta)^2$. In the notation preceding Lemma 7.4 define the nonnegative endpoint measure $m_n = (|\lambda_n^+| + |\lambda_n^-|)/2$. Both $|\sigma_n|$ and $|\tau_n|$ are bounded pointwise by m_n , and $\|m_n\|_1 = M_n$. The lemma gives

$$\|Q\bar{h}\|_H^2 + \|\tilde{h}\|_H^2 + \sum_n d_n^2 \leq (R + \eta)^2 - \|x\|_q^2 =: E. \quad (68)$$

Write $S_t = \{s : s \succeq t\}$ for a descendant set. Mark a node $t \notin A_n$ when $|\sigma_n(S_t)| < m_n(S_t)/2$, and retain only the first marked nodes on each path outside the head. Every marked node has such a first ancestor, since it has finitely many ancestors. The retained descendant sets are pairwise disjoint.

Each retained set lies in a single fiber of r . If disjoint subsets S_j lie in a nondiscarded fiber C , then

$$\sum_j (m_n(S_j) - |\sigma_n(S_j)|) \leq m_n(C) - |\sigma_n(C)|.$$

Indeed $|\sigma_n(C)|$ is at most the sum of the displayed absolute sums plus the m_n -mass of the remaining set. This argument also holds for countably many subsets by absolute convergence. Discarded fibers are bounded by their whole m_n -mass. Summing over fibers yields

$$\sum_{t \text{ first marked}} (m_n(S_t) - |\sigma_n(S_t)|) \leq d_n, \quad \sum_{t \text{ first marked}} m_n(S_t) \leq 2d_n.$$

Split $\tau_n = \tau_n^r + \tau_n^k$ by restricting the first part to the union of these descendant sets. Then $\|\tau_n^r\|_1 \leq 2d_n$ and

$$\|QJ\tau^r\|_q \leq 4\left(\sum_n d_n^2\right)^{1/2}.$$

If an outside node has a first marked ancestor, the occupancy of τ_n^k there is zero. Otherwise it is unmarked, and

$$|(J_n\tau_n^k)(t)| \leq m_n(S_t) \leq 2|\sigma_n(S_t)| = 2|\bar{h}_n(t)|.$$

The last equality uses $Qx = 0$ in $x = \bar{h} + J\sigma$. Thus $QJ\tau^k \in H$ and has Hilbert norm at most $2\|Q\bar{h}\|_H$. Since $y = \tilde{h} + J\tau$,

$$\|Qy\|_q \leq \|Q\tilde{h}\|_H + 2\|Q\bar{h}\|_H + 4\left(\sum_n d_n^2\right)^{1/2} \leq 7\sqrt{E}.$$

Let $\eta \downarrow 0$. □

Corollary 7.11. *For $0 < c \leq 1$ put $\delta = c^2/2000$. If $\|x \pm y_j\|_q \leq 1$ and $\|y_i - y_j\|_q \geq 2c$ for all distinct indices of an infinite sequence, then $\|x\|_q \leq 1 - \delta$.*

Proof. If $\|x\|_q > 1 - \delta$, choose a finite-support x_0 within δ of x , and an ancestral head A for x_0 . Then $\|x_0\|_q > 1 - 2\delta$ and $\|x_0 \pm y_j\|_q \leq 1 + \delta$. Proposition 7.10 gives

$$\|Q_A y_j\|_q < 7\sqrt{(1 + \delta)^2 - (1 - 2\delta)^2} \leq 7\sqrt{6\delta} < c/2.$$

Two bounded head projections are less than c apart, contradicting the assumed separation of the full vectors. □

7.6 Pruning by a majority sign

This argument tracks the labeled majority terms before cancellation in the half-difference. It keeps those terms until their occupancy is comparable with the minority occupancy, separating removed mass from a remaining occupancy bounded in Hilbert norm. This decomposition will also be used after approximating an arbitrary center.

Proposition 7.12. *Under the hypotheses of Theorem 7.6,*

$$\|Q_A y\|_q \leq 9\sqrt{R^2 - \|x\|_q^2}.$$

Proof. Use representations with squared costs at most $R^2 + \eta$ and write $E = R^2 + \eta - \|x\|_q^2$. In each nondiscarded fiber C_r , orient the coefficients by the sign of the folded coefficient $\nu_n(r)$, choosing +1 when it is zero; use either sign in $\{-1, 1\}$ in a discarded fiber. Let g_n, b_n be half the total absolute masses of coefficients agreeing and disagreeing with this sign, respectively, counting both labeled endpoint representations. Then $\|b_n\|_1 \leq d_n$. More precisely, if $A_n \neq \emptyset$, all nodes have an ancestor in A_n and $2\|b_n\|_1 = d_n$; when $A_n = \emptyset$, simply $d_n = M_n$.

On the outside put $G(t) = g_n(S_t)$ and $B(t) = b_n(S_t)$. Constancy of the orientation along each outside subtree gives

$$|(J_n \sigma_n)(t)| = |G(t) - B(t)| = |\bar{h}_n(t)|.$$

In the labeled half-difference $\tau_n = (\lambda_n^+ - \lambda_n^-)/2$, remove every minority term and every majority term whose endpoint has an outside ancestor t with $G(t) \leq 2B(t)$. Denote the removed and kept arrays by τ_n^r, τ_n^k . The first such stopping nodes have disjoint descendant sets. The removed majority mass is at most $2\|b_n\|_1$; hence $\|\tau_n^r\|_1 \leq 3d_n$. This accounting is on the labeled endpoint terms, before any cancellation in their half-difference, so it bounds that difference as well.

At a node having a stopping ancestor the kept occupancy vanishes. At every other outside node, $G(t) > 2B(t)$ and

$$|(J_n \tau_n^k)(t)| \leq G(t) \leq 2|G(t) - B(t)| = 2|\bar{h}_n(t)|.$$

Therefore

$$\|Q y_n\|_n \leq \|Q \tilde{h}_n\|_2 + 2\|Q \bar{h}_n\|_2 + 6d_n.$$

Lemma 7.4 and the triangle inequality in the outer Euclidean sum give $\|Q y\|_q \leq 9\sqrt{E}$. Let $\eta \downarrow 0$. $\square$

Corollary 7.13. *For every $\rho > 0$, choose $0 < \gamma < 1/2$ such that*

$$9\sqrt{(1 + \gamma)^2 - (1 - 2\gamma)^2} < \rho/3.$$

If $\|x\|_q \geq 1 - \gamma$, its symmetric lens $\{y : \|x \pm y\|_q \leq 1\}$ has no infinite ρ -separated subset.

Proof. Approximate x within γ by a finite-support x_0 , and choose a finite ancestral head for it. The tail of every point in the lens has norm less than $\rho/3$ by Proposition 7.12. The points in the lens have norm at most one, so in an infinite sequence two head projections are less than $\rho/3$ apart. Their full distance is less than ρ . $\square$

The next formulation chooses one head from a nonzero center and a relative endpoint excess before any displacement is supplied. It therefore controls every admissible displacement at that center with the same head. This is the form needed when several displacements are compared with one center.

Proposition 7.14. *For every nonzero $x \in X_q$ and $0 < \delta \leq 1$ there is a finite ancestral head A , depending only on x, δ , such that for every $u \in X_q$,*

$$\|x + u\|_q, \|x - u\|_q \leq (1 + \delta)\|x\|_q \implies \|Q_A u\|_q \leq 72\sqrt{\delta}\|x\|_q.$$

Proof. Scale to $\|x\|_q = 1$. Choose x_0 of finite node support with $\|x - x_0\|_q \leq \delta$ and an ancestral head A containing its support. Then $\|x_0 \pm u\|_q \leq 1 + 2\delta$. Choose representations $x_0 \pm u = h^\pm + J\lambda^\pm$ with each squared cost at most $(1 + 3\delta)^2$. Using the notation above for these representations, let C be their average squared cost and set

$$\mathcal{D} = C - \left(\|P\bar{h}\|_H^2 + \sum_n q_n^2 \right).$$

The parallelogram identity and the scalar square inequality used in Lemma 7.4 give, before comparing the center representation with its norm,

$$\begin{aligned} \|Q\bar{h}\|_H^2 + \|\tilde{h}\|_H^2 + \sum_n (M_n^2 - q_n^2) &\leq \mathcal{D}, \\ \frac{1}{2}(\|Qh^+\|_H^2 + \|Qh^-\|_H^2) + \sum_n d_n^2 &\leq \mathcal{D}. \end{aligned}$$

The second inequality uses $M_n^2 - q_n^2 \geq d_n^2$ and the parallelogram identity for $Qh^\pm$. The center representation has cost at least $\|x_0\|_q$, so

$$0 \leq \mathcal{D} \leq (1 + 3\delta)^2 - \|x_0\|_q^2 \leq (1 + 3\delta)^2 - (1 - \delta)^2 \leq 16\delta. \quad (69)$$

Orient each nondiscarded fiber C_r by the sign of $\nu_n(r)$, using $+1$ for a zero folded coefficient or a discarded fiber. This time use the full, rather than half, outside absolute masses: let $p_n^\pm$ agree with the orientation and $b_n^\pm$ disagree with it, and put $B_n = \|b_n^+ + b_n^-\|_1$. The outside minority mass is at most the total cancellation bound, giving $B_n \leq 2d_n$ and

$$\sum_n B_n^2 \leq 4\mathcal{D}. \quad (70)$$

For outside nodes let $G(t) = (p_n^+ + p_n^-)(S_t)$ and $V(t) = (b_n^+ + b_n^-)(S_t)$. Since the averaged represented vector vanishes there,

$$|G(t) - V(t)| = |h_n^+(t) + h_n^-(t)|.$$

Stop at the first outside nodes satisfying $G(t) > 0$ and $V(t) \geq G(t)/2$. Their descendant sets are disjoint, and the total majority mass in them is at most $2B_n$. Remove all these majority terms and all minority terms from the outside part of λ_n^k ; denote what remains by λ_n^r and what was removed by λ_n^k . Then $\|\lambda_n^r\|_1 \leq 3B_n$ and, by (70),

$$\|\lambda^r\|_L \leq 6\sqrt{\mathcal{D}}, \quad \|QJ\lambda^r\|_q \leq 12\sqrt{\mathcal{D}}.$$

Every outside node either has a stopping ancestor, has $G(t) = 0$, or satisfies $V(t) < G(t)/2$. In all cases the kept occupancy is at most $2|G(t) - V(t)|$ in absolute value. Thus

$$\|QJ\lambda^k\|_H \leq 2\|Q(h^+ + h^-)\|_H \leq 4\sqrt{\mathcal{D}}.$$

The atomic terms whose endpoints belong to A have zero tail. The representation of $x_0 + u$, together with $\|Qh^+\|_H \leq \sqrt{2\mathcal{D}}$, now gives

$$\|Qu\|_q \leq (12 + 4 + \sqrt{2})\sqrt{\mathcal{D}} \leq 18\sqrt{\mathcal{D}} \leq 72\sqrt{\delta}.$$

All stopping decisions use a first ancestor, so infinite coefficient support and countably many children cause no change. The choice of A preceded the choice of u , as required. $\square$

7.7 From a component estimate to the Hilbert sum

We now obtain a finite-codimensional midpoint estimate from normalized component bounds. The proof applies a local lens estimate on components with controlled relative endpoint excess and estimates the remaining components by their contribution to the total squared excess.

Lemma 7.15 (Component estimate). *Let $A \subset T_n$ be a finite nonempty ancestral head, with $P_A x = x$, $P_A y = 0$, and $|x|_n = 1$. If $0 \leq e \leq 1$ and $|x \pm y|_n \leq 1 + e$, then*

$$|y|_n \leq (6 + \sqrt{2})\sqrt{(1 + e)^2 - 1} \leq 13\sqrt{e}. \quad (71)$$

Proof. Apply the proof of Theorem 7.6 in this single component, retaining the coefficient $6 + \sqrt{2}$ before its final rounding to 8. Here $Q_A y = y$. Explicitly, approximate endpoint representations of cost below $1 + e + \varepsilon$ give $\kappa = (1 + e + \varepsilon)^2 - 1$, Hilbert tails at most $\sqrt{2\kappa}$, combined Hilbert tail at most $2\sqrt{\kappa}$, and folded-mass loss at most $\sqrt{\kappa}$ by Lemma 7.4. Because the head contains the root, no fiber is discarded: the combined minority mass is exactly this folded-mass loss. Grouping by the common truncated endpoint and clipping one majority occupancy against the combined minority occupancy gives two signed stopped flows of norm at most $2\sqrt{\kappa}$ each, by Lemma 7.5. The remaining majority occupancy is bounded in Hilbert norm by $2\sqrt{\kappa}$. Adding the original Hilbert tail gives $(6 + \sqrt{2})\sqrt{\kappa}$. These are the same decomposition and signed path representations used in that theorem, so no invariance of $|\cdot|_n$ under arbitrary coordinate sign changes is needed. Let $\varepsilon \downarrow 0$. Finally $(1 + e)^2 - 1 \leq 3e$ and $3(6 + \sqrt{2})^2 = 114 + 36\sqrt{2} < 169$. $\square$

The component estimate controls a lens after normalization by the center norm. In a Hilbert sum that norm can be very small or zero. The next argument separates the components carrying a large relative endpoint excess before making any such division.

Proposition 7.16 (Finite centers in the Hilbert sum). *Let $x \in X_q$ be a unit vector with finite coordinate support. For each n with $x_n \neq 0$, choose a finite nonempty ancestor-closed $A_n \subset T_n$ containing its support, and set*

$$Y_x = \{y \in X_q : P_{A_n} y_n = 0 \text{ whenever } x_n \neq 0\}.$$

Then Y_x is a closed finite-codimensional subspace. If $0 < e \leq 1/8$, $y \in Y_x$, and $\|x + y\|_q, \|x - y\|_q \leq 1 + e$, then

$$\|y\|_q^2 \leq \frac{519}{2}e^{1/3}, \quad \|y\|_q \leq 17e^{1/6}. \quad (72)$$

Proof. The conditions defining Y_x involve only finitely many continuous coordinate functionals, so the assertion about codimension follows. Put

$$a_n = |x_n|_n, \quad b_n = |x_n + y_n|_n, \quad c_n = |x_n - y_n|_n, \quad v_n = \frac{b_n^2 + c_n^2}{2}, \quad d_n = v_n - a_n^2.$$

The triangle inequality gives $a_n, |y_n|_n \leq (b_n + c_n)/2$. Therefore

$$d_n \geq 0, \quad |y_n|_n^2 \leq v_n, \quad \sum_n a_n^2 = 1, \quad \sum_n d_n \leq (1 + e)^2 - 1 \leq 3e. \quad (73)$$

Fix $0 < \eta \leq 1$. Write

$$E_\eta = \{n : d_n > \eta a_n^2\}.$$

On these components $v_n = a_n^2 + d_n \leq (1 + 1/\eta)d_n$, and hence

$$\sum_{n \in E_\eta} |y_n|_n^2 \leq 3e(1 + 1/\eta). \quad (74)$$

If $a_n = 0$ and $y_n \neq 0$, then $d_n = v_n = |y_n|_n^2 > 0$, so $n \in E_\eta$. If both vectors vanish in that component, its contribution is zero. Thus every remaining nonzero contribution has $a_n > 0$.

Now assume additionally that $\eta \leq 1/4$. For $n \notin E_\eta$ with $a_n > 0$, let $m_n = (b_n + c_n)/2$. We have $m_n \geq a_n$ and

$$m_n^2 + \left(\frac{b_n - c_n}{2}\right)^2 = v_n \leq (1 + \eta)a_n^2.$$

It follows that

$$\frac{\max(b_n, c_n)}{a_n} \leq \sqrt{1 + \eta} + \sqrt{\eta} = 1 + \delta_\eta, \quad \delta_\eta := \sqrt{1 + \eta} + \sqrt{\eta} - 1 \leq \frac{3}{2}\sqrt{\eta} \leq 1.$$

Apply Lemma 7.15 to x_n/a_n and y_n/a_n . Summing its squared estimate gives

$$\sum_{n \notin E_\eta} |y_n|_n^2 \leq 169\delta_\eta \sum_{n \notin E_\eta} a_n^2 \leq \frac{507}{2}\sqrt{\eta}. \quad (75)$$

Take $\eta = e^{2/3}$, which is at most $1/4$ when $e \leq 1/8$. Combining (74) and (75),

$$\|y\|_q^2 \leq \frac{507}{2}e^{1/3} + 3e + 3e^{1/3} \leq \frac{519}{2}e^{1/3}.$$

The second assertion follows from $519/2 < 17^2$. $\square$

Corollary 7.17 (Completed centers). *For every $x \in X_q$ with $\|x\|_q = 1$ and $0 < e \leq 1/16$, there is a closed finite-codimensional subspace $Y \subset X_q$ such that*

$$y \in Y, \quad \|x + y\|_q, \|x - y\|_q \leq 1 + e \quad \implies \quad \|y\|_q \leq 17(2e)^{1/6}. \quad (76)$$

In particular, for every $t > 0$ there is an $e_t > 0$, independent of the unit center x , for which some closed finite-codimensional Y satisfies

$$\inf_{u \in S_Y} \max\{\|x + tu\|_q, \|x - tu\|_q\} \geq 1 + e_t.$$

Here $S_Y = \{u \in Y : \|u\|_q = 1\}$. Thus the maximum asymptotic midpoint modulus is positive at every positive radius.

Proof. By finite-coordinate density, choose a finitely supported unit vector x^0 with $\|x - x^0\|_q < e$. Such unit vectors are dense: approximate x by a nonzero finitely supported vector and normalize it. Use the finite coordinate exclusions for x^0 from Proposition 7.16. If y satisfies the hypotheses of (76), then

$$\|x^0 + y\|_q, \|x^0 - y\|_q \leq 1 + 2e.$$

Since $2e \leq 1/8$, Proposition 7.16 proves (76). Given $t > 0$, choose $0 < e_t \leq 1/16$ with $17(2e_t)^{1/6} < t$. For $u \in S_Y$, both endpoint norms at tu cannot be at most $1 + e_t$, by (76). Each corresponding maximum is therefore greater than $1 + e_t$, giving the displayed infimum bound. $\square$

8 Charging segments at their starting nodes

We now allow an atom to start at any node of a tree. Its cost depends on where it starts: masses on one branch add, whereas masses entering distinct child subtrees combine in Euclidean norm. The resulting quotient norm has a squared midpoint estimate. The proof matches opposite signed segments at their first node outside a finite head and charges the discarded part to the later of their two starting nodes.

8.1 The cost and the completed space

For each integer $n \geq 1$, let $\mathcal{T}_n = \mathbb{N}^{\leq n}$, including its root $\emptyset$. Give nodes their component label, so the trees are disjoint. Write $t \preceq v$ when t is an initial segment of v , and $[t, v] = \{s : t \preceq s \preceq v\}$. All scalars in this section are real. For a nonnegative array m on $\mathcal{T}_n$, define, upwards from the leaves,

$$q_t(m) = m_t + \left(\sum_{s \text{ child of } t} q_s(m)^2 \right)^{1/2}, \quad M_n(m) = q_\emptyset(m). \quad (77)$$

Empty sums are zero; infinite values are allowed in this definition. On the disjoint union $\mathcal{T}$ of the trees put

$$M(m) = \left(\sum_{n \geq 1} M_n(m|_{\mathcal{T}_n})^2 \right)^{1/2}.$$

Thus M_n is a cost for nonnegative start masses, before any vector norm has been defined.

Lemma 8.1. *The costs are monotone, positively homogeneous and subadditive. On $\mathcal{T}_n$ they satisfy*

$$\|m\|_2 \leq M_n(m) \leq \sum_{d=0}^n \|m|_{\mathbb{N}^d}\|_2 \leq \sqrt{n+1} \|m\|_2, \quad \sum_{t \preceq v} m_t \leq M_n(m). \quad (78)$$

For nonnegative forest arrays of finite total cost,

$$M(m+r)^2 \geq M(m)^2 + M(r)^2. \quad (79)$$

Proof. Monotonicity and homogeneity follow directly by induction through the levels. The same induction and the triangle inequality in ℓ_2 give subadditivity. Squaring (77) and discarding its nonnegative cross term proves the lower bound in (78). The ℓ_2 norm of the q_t on level d is at most the ℓ_2 norm of the m_t on that level plus the ℓ_2 norm of the q_s on level $d+1$. Iterating gives the middle bound; Cauchy–Schwarz gives the last one. Keeping only the child on the path to v at each step gives the path bound.

For the energy inequality, induction proves $q_t(m+r) \geq (q_t(m)^2 + q_t(r)^2)^{1/2}$. Indeed, let A and B be the ℓ_2 norms of the child values for m and r . The induction hypothesis gives

$$q_t(m+r) \geq m_t + r_t + \sqrt{A^2 + B^2} \geq \sqrt{m_t^2 + r_t^2} + \sqrt{A^2 + B^2} \geq \sqrt{(m_t + A)^2 + (r_t + B)^2}.$$

The last inequality is the triangle inequality in $\mathbb{R}^2$. Summing its square over the roots proves (79). These arguments apply to countably many children: all sums are nonnegative, and the ℓ_2 inequalities hold for finite partial sums and then by monotone convergence. This also shows that each cost commutes with increasing limits of nonnegative arrays. $\square$

A segment representation on $\mathcal{T}_n$ is an array $a = (a_{tv})_{t \preceq v}$ with start masses

$$m_t(a) = \sum_{v \succeq t} |a_{tv}| \quad \text{and} \quad M_n(m(a)) < \infty.$$

Its output is

$$(Pa)_s = \sum_{t \preceq s} \sum_{v \succeq s} a_{tv}. \quad (80)$$

Each coefficient a_{tv} contributes its value on the entire segment $[t, v]$. The sums at a fixed coordinate converge absolutely, by the path bound in (78).

Proposition 8.2. *On $\ell_2(\mathcal{T}_n)$ the formula*

$$\|x\|_{S,n} = \inf\{M_n(m(a)) : Pa = x\}$$

defines a complete reflexive norm, with

$$\frac{\|x\|_2}{n+1} \leq \|x\|_{S,n} \leq \sqrt{n+1} \|x\|_2. \quad (81)$$

Consequently

$$X_S = \left(\bigoplus_{n \geq 1} (\ell_2(\mathcal{T}_n), \|\cdot\|_{S,n}) \right)_{\ell_2}$$

is a reflexive Banach space. Write $\|\cdot\|_S$ for its norm. Equivalently, $\|x\|_S$ is the infimum of $M(m(a))$ over all forest representations of x . Finite-support vectors are dense.

Proof. First we bound the output of a representation. Fix an output level d and a start level $h \leq d$. Segments starting at a single node t of level h contribute total absolute mass at most $m_t(a)$ to level d ; starts at different nodes of level h have disjoint descendants. Therefore the ℓ_2 norm of this part of the output is at most $\|m(a)|_{\mathbb{N}^h}\|_2$. Summing over $h \leq d$ bounds the norm on level d by $\sqrt{n+1} M_n(m(a))$. Summing the squares over d gives

$$\|Pa\|_2 \leq (n+1) M_n(m(a)). \quad (82)$$

Conversely, the singleton representation $a_{tt} = x_t$ has cost at most $\sqrt{n+1} \|x\|_2$. This proves finiteness and the comparisons. Homogeneity is immediate. Adding representations and using monotonicity and subadditivity of the cost proves the triangle inequality; the lower comparison proves definiteness. The norm is equivalent to the Hilbert norm, and hence is complete and reflexive.

The countable ℓ_2 sum is complete. Its dual is isometrically the ℓ_2 sum of the component duals: restriction of a functional gives the component functionals, finite collections of approximate norming vectors give the lower bound for their ℓ_2 norm, and Cauchy–Schwarz gives the reverse bound. Finite-component vectors are dense, so restrictions determine the functional. Applying this identification twice and using the reflexivity of every component proves reflexivity of X_S .

A forest representation bounds each component norm by its component cost, and therefore bounds $\|x\|_S$ by its total cost. Conversely, for $\eta > 0$ choose a representation in component n with cost at most $\|x|_{\mathcal{T}_n}\|_{S,n} + \eta 2^{-n}$. Minkowski's inequality makes its total cost less than $\|x\|_S + \eta$. Finally, finite support is dense in each component by (81), and finite-component vectors are dense in the sum. $\square$

We will keep separate entries on identical segments when convenient. More precisely, one may use a countable list of signed segment weights whose absolute start masses have finite cost. Combining entries on each segment only decreases the start masses and gives a representation in the preceding sense. Splitting a weight into pieces of the same sign preserves the masses. These observations allow the matching argument below without any assumption that an infimum is attained.

A subset of $\mathcal{T}$ is *ancestral* if it contains every ancestor of each of its nodes.

Lemma 8.3. *For every node v and $x \in X_S$, $|x_v| \leq \|x\|_S$. The coordinate vector e_v and every root-to-node path vector have norm one. If C is a finite ancestral subset of $\mathcal{T}$, then its coordinate projection P_C is contractive and $\|I - P_C\| \leq 2$. For a nonterminal node t , the sequence of its child coordinate vectors is weakly null.*

Proof. The absolute value of the output at v is at most the sum of start masses on the ancestor path to v , which is bounded by $M(m(a))$. Take the infimum over representations. A singleton or a root-to-node path has a representation of cost one, and its nonzero coordinate gives the matching lower bound.

Restrict each representing segment to C . Since C contains every ancestor of its nodes, this restriction is empty or a segment with the same start. The start masses cannot increase. Taking infima proves $\|P_C x\|_S \leq \|x\|_S$; the bound for $I - P_C$ follows by the triangle inequality. The child vectors lie in a fixed component and are weakly null there, because its norm is equivalent to the Hilbert norm. They remain weakly null under its inclusion in X_S . $\square$

The unit path vectors in arbitrarily tall trees, together with their weakly null unit increments, satisfy the hypotheses of Lemma 2.3. Thus X_S admits no equivalent asymptotically uniformly convex norm. More explicitly, if $\alpha\|x\|_S \leq N(x) \leq \beta\|x\|_S$ and the modulus of N were positive at $t_0 = \alpha/(2\beta)$, the lemma would extend a path with a factor $1 + \bar{\delta}_N(t_0)/4$ at every step. Its N norm would eventually exceed β , although every such path has $\|\cdot\|_S$ norm one.

8.2 A squared estimate across a finite cut

The next theorem is the geometric estimate supplied by this model. Unlike a root-path cost, the present cost records the starting node of each segment. Removing one of two opposite crossing segments saves the mass charged at its later start. The positive energy inequality then turns that saved mass into a squared gain.

Theorem 8.4. *Let C be a finite ancestral subset of $\mathcal{T}$. If $p \in X_S$ is supported in C and $y \in X_S$ satisfies $y|_C = 0$, then*

$$\max\{\|p + y\|_S, \|p - y\|_S\}^2 \geq \|p\|_S^2 + \frac{\|y\|_S^2}{64}. \quad (83)$$

The conclusion includes $C = \emptyset$, $p = 0$ and $y = 0$.

Proof. Fix $\eta > 0$ and put $R = \max\{\|p + y\|_S, \|p - y\|_S\} + \eta$. Choose representations of $p + y$ and $p - y$ of cost at most R . Halve every coefficient and concatenate the two lists, retaining the labels $+$ and $-$ that record their origins. These labels are independent of the signs of the coefficients. The combined list has output p and start mass m with $M(m) \leq R$. Figure 1 illustrates the two uses of a matched pair.

A first node outside C means a node $s \notin C$ whose parent lies in C , or a root outside C . Every segment is then of exactly one of three kinds: it lies wholly in C ; it contains one first node outside C ; or its start lies strictly below a first node outside C . The second kind includes segments starting at

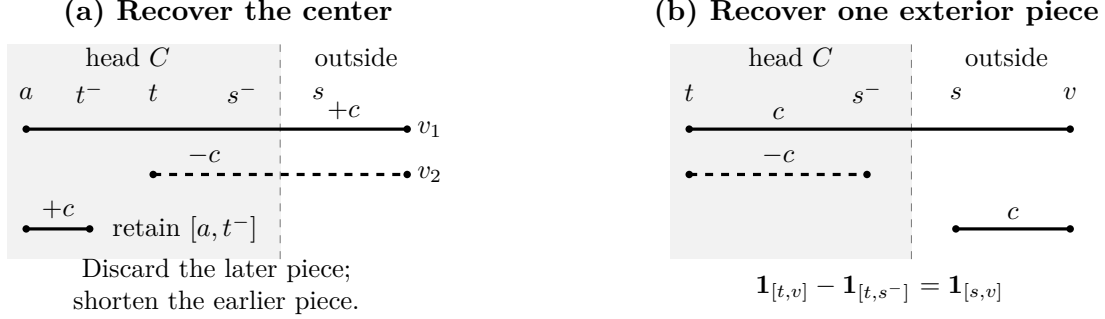


Figure 1: Segment matching in the representative case $a \prec t \preceq s^-$, where a and t are the earlier and later starts and s is the first node outside the head. Each row represents a segment; its labels through s refer to the common ancestor chain. In (a), the displayed signs are opposite coefficient signs, with $c > 0$; they are independent of the $+$ and $-$ endpoint-origin labels. In (b), a piece from the $+$ endpoint can have either sign of c . Its exterior is reproduced using at most two segments starting at t . The proof also covers coincident starts, $t = s$, and roots outside C .

that first node. At each such s , match the positive and negative weights of all segments containing s in pieces of equal absolute weight. Their total positive and total negative masses are finite and equal, since their sum is the output $p_s = 0$. For completeness, place the positive weights consecutively on an interval of their total length, do the same with the absolute negative weights, and use the intersections of these intervals as matching pieces. This produces a countable matching that uses every weight. A segment contains at most one first node outside C , so the matchings do not interfere.

In each matched pair, both starts are ancestors of its first outside node s . Call the piece with deeper start the later piece, breaking a tie arbitrarily. Define r_t to be the sum of the sizes of later pieces starting at t , together with the sizes of all original pieces starting strictly below a first node outside C . These are disjoint portions of the list; hence $0 \leq r \leq m$.

We first recover p at cost at most $M(m - r)$. Delete all pieces starting strictly below a first outside node. For a matched pair with coincident starts, discard both pieces. Otherwise discard the later piece and shorten the earlier piece to end immediately before the later start. The remaining part lies in C . On C the removed overlapping parts had equal opposite coefficients and therefore canceled. Segments wholly inside C are unchanged. The modified list consequently represents p , and its start masses are at most $m - r$. By (79),

$$\|p\|_S^2 + M(r)^2 \leq M(m - r)^2 + M(r)^2 \leq M(m)^2. \quad (84)$$

It remains to control y by the saved mass. The output of the $+$ list outside C is $y/2$. We represent it using start masses at most $4r$. Include each $+$ piece starting strictly below a first outside node without change. For a $+$ piece in a pair at s , let t be the pair's later start and v the endpoint of this piece. Replace its exterior part by the segment $[t, v]$ with its original coefficient. If $t \in C$, also include the segment from t to the parent of s with the opposite coefficient. These two segments leave precisely the desired output from s to v . If $t = s$, no subtraction is needed. Each pair has at most two $+$ pieces, and each uses at most two segments of the matched size, all starting at t . Their total start mass is therefore at most four times the later piece's mass. No segment wholly in C is needed. Thus

$$\|y\|_S \leq 2M(4r) = 8M(r). \quad (85)$$

All the modified lists are legitimate: their absolute start masses have finite cost, and each coordinate has only finitely many ancestors. Combining (84) and (85) gives $\|p\|_S^2 + \|y\|_S^2/64 \leq R^2$.

Finally let $\eta \downarrow 0$. □

Corollary 8.5. *For every $x \in X_S$ with $\|x\|_S = 1$, every $t > 0$ and every $\varepsilon > 0$ there is a closed finite-codimensional subspace $F \subset X_S$ such that*

$$\inf_{\substack{y \in F \\ \|y\|_S = 1}} \max\{\|x + ty\|_S, \|x - ty\|_S\} \geq \sqrt{1 + t^2/64} - \varepsilon.$$

In particular, define the maximum form of the asymptotic midpoint modulus by

$$\delta_{X_S}^{\max}(t) = \inf_{\|x\|_S = 1} \sup_{\substack{F \subset X_S \text{ closed} \\ \dim(X_S/F) < \infty}} \inf_{\substack{y \in F \\ \|y\|_S \geq 1}} (\max_{\pm} \|x \pm ty\|_S - 1).$$

Then $\delta_{X_S}^{\max}(t) \geq \sqrt{1 + t^2/64} - 1$.

Proof. Choose a finite-support p arbitrarily close to x , let C be its finite ancestral closure and set $F = \ker P_C$. Theorem 8.4 gives, uniformly for $y \in F$ with $\|y\|_S \geq 1$,

$$\max_{\pm} \|x \pm ty\|_S \geq \sqrt{\|p\|_S^2 + t^2/64} - \|x - p\|_S.$$

As $\|x - p\|_S \rightarrow 0$, the right side tends to $\sqrt{1 + t^2/64}$. Coordinate continuity makes F closed and finite-codimensional, proving the assertion. □

9 Incomparable segments

In this section, arrays are tested by families of segments in which every node of one segment is incomparable with every node of another. The resulting dual space has unit root paths of arbitrarily large finite height. When both endpoints lie in the unit ball, a midpoint close to the unit sphere forces small displacement beyond any finite prefix supporting that midpoint. We prove the latter assertion by replacing the coefficients of segment atoms with the sums of a norming test along their segments. Opposite signs can then be controlled by shortening segments without violating their incomparability.

9.1 The norm and its dual atoms

For an integer $h \geq 1$, let $T_h = \mathbb{N}^{\leq h}$, including the empty sequence as its root, and take disjoint copies when forming $\mathcal{T} = \bigsqcup_{h \geq 1} T_h$. Nodes in the same tree are ordered by extension, denoted by $s \preceq t$; nodes in different trees are incomparable. If $s \preceq t$, the segment $[s, t]$ consists of every node between s and t , including both endpoints. A finite family $(S_i)_{i=1}^m$ of nonempty segments is *incomparable* if, for $i \neq j$, every node of S_i is incomparable with every node of S_j . In particular, the segments are disjoint, but disjointness alone is not this condition.

For a finitely supported real array g on $\mathcal{T}$, set

$$\|g\|_{Y_I} = \sup_{(S_i) \text{ incomparable}} \left(\sum_{i=1}^m \left| \sum_{v \in S_i} g(v) \right|^2 \right)^{1/2}, \quad (86)$$

and let Y_I be its completion. This is a finite norm on $c_{00}(\mathcal{T})$: the triangle inequality follows from the Euclidean triangle inequality, singleton tests separate points, and disjointness bounds every

test by $\sum_v |g(v)|$. Each coordinate and each finite segment evaluation extends continuously to Y_I . Formula (86) remains valid in the completion. Indeed, the supremum of the extended tests is at most the completion norm and is Lipschitz with constant one; approximation by finitely supported arrays gives the reverse inequality.

Quadratic aggregation over incomparable segment families also appears in the ℓ_2 -Baire-sum construction of Argyros and Dodos [1, author version, Section 4.1]. There each segment contributes the norm of a sum in a Schauder tree basis; here it contributes the absolute scalar sum in (86).

Proposition 9.1. *Let $Y_{I,h}$ be the completion of the arrays supported on T_h in the norm (86). Then*

$$\frac{\|g\|_2}{\sqrt{h+1}} \leq \|g\|_{Y_{I,h}} \leq \sqrt{h+1} \|g\|_2, \quad g \in c_{00}(T_h), \quad (87)$$

and

$$Y_I = \left(\bigoplus_{h \geq 1} Y_{I,h} \right)_{\ell_2}.$$

Both Y_I and $X_I := Y_I^*$ are reflexive.

Proof. At each fixed level of T_h , the singleton segments form an incomparable family. Hence the squared norm of g dominates the sum of $|g(v)|^2$ over that level. Summing over the $h+1$ levels gives the lower bound in (87). For the upper bound, every segment has at most $h+1$ nodes, so Cauchy–Schwarz and disjointness give

$$\sum_i \left| \sum_{v \in S_i} g(v) \right|^2 \leq (h+1) \sum_i \sum_{v \in S_i} |g(v)|^2 \leq (h+1) \|g\|_2^2.$$

Thus $Y_{I,h}$ is isomorphic to the Hilbert space $\ell_2(T_h)$ and is reflexive.

A test family separates into families on the individual trees. Conversely, the union of finitely many such families, one in each of distinct trees, is incomparable. Choosing each component family arbitrarily close to its supremum proves

$$\|g\|_{Y_I}^2 = \sum_h \|g|_{T_h}\|_{Y_{I,h}}^2$$

for finitely supported g , and then the asserted sum decomposition follows by completion. For completeness, the dual of an ℓ_2 -sum of Banach spaces is the ℓ_2 -sum of their duals: Cauchy–Schwarz bounds the sum of the pairings, while finitely many nearly norming component vectors with suitable scalar weights give the opposite norm inequality. Restrictions to the components determine a functional by density. Applying this description twice, componentwise reflexivity shows that the canonical map into the bidual is onto. Thus Y_I is reflexive, as is its dual X_I . $\square$

Write $e_v \in X_I$ for evaluation at v , and put $z_S = \sum_{v \in S} e_v$, with $z_\emptyset = 0$. We denote the dual norm by $\|\cdot\|_I$. The coordinate evaluations are linearly independent, so X_I is infinite dimensional. An *atom* is a vector together with a representation

$$A = \sum_{i=1}^m a_i z_{S_i}, \quad \sum_{i=1}^m a_i^2 \leq 1, \quad (S_i)_{i=1}^m \text{ incomparable.} \quad (88)$$

The representation will matter when its coefficients are replaced below.

Lemma 9.2. *The unit ball of X_I is the norm-closed convex hull of the atoms (88). In particular, finite linear combinations of the coordinate evaluations are dense in X_I .*

Proof. Every atom has norm at most one, by Cauchy–Schwarz and (86), and for every $g \in Y_I$,

$$\sup_{A \text{ atom}} A(g) = \|g\|_{Y_I}.$$

If the closed convex hull of the atoms missed a point of the dual unit ball, Hahn–Banach separation would give a separating functional in $X_I^* = Y_I$, using Proposition 9.1. The displayed identity rules out such a separation. Each atom has finite support, which gives the density assertion. $\square$

9.2 Unit paths and finite prefixes

The root paths give the one-sided obstruction; finite prefixes will separate the head of a midpoint from its remaining coordinates. For $s \in T_h$, define

$$v_s = z[\varnothing_h, s],$$

where $\varnothing_h$ denotes the root of that copy of T_h . Each v_s is an atom of norm one: its norm is at most one by Lemma 9.2, and testing it against the array which equals one at s and zero elsewhere gives the reverse inequality. The same argument gives $\|e_t\|_I = 1$ for every node t . If $|s| < h$ and $t_j = s \frown j$, then

$$v_{t_j} = v_s + e_{t_j}, \quad \|v_s\|_I = \|e_{t_j}\|_I = 1, \quad e_{t_j} \longrightarrow 0 \text{ weakly.} \quad (89)$$

To verify weak convergence, singleton tests at the children show that $\sum_j |g(t_j)|^2 \leq \|g\|_{Y_I}^2$ for each $g \in Y_I$. Thus $g(t_j) \rightarrow 0$, and these are all the functionals on X_I by reflexivity.

Corollary 9.3. *The real reflexive space X_I admits no equivalent asymptotically uniformly convex norm.*

Proof. The trees (89) exist at every finite height and satisfy Lemma 2.3 with path bounds and increment lower bound all equal to one. More explicitly, for an equivalent norm N with $a\|u\|_I \leq N(u) \leq b\|u\|_I$, that lemma uses $t_0 = a/(2b)$. If $\delta = \bar{\delta}_N(t_0) > 0$, its finite-codimensional approximation and convexity argument selects a child at every nonterminal node for which

$$N(v_{t_j}) \geq (1 + \delta/4)N(v_s).$$

Iteration to height h gives $a(1 + \delta/4)^h \leq b$, a contradiction for sufficiently large h . $\square$

A set $E \subset \mathcal{T}$ is *prefix closed* if it contains every ancestor of each of its nodes, within that node’s tree. Let P_E be coordinate restriction to E , and let $Q_E = I - P_E$.

Lemma 9.4. *For every finite prefix-closed E , both P_E and Q_E act contractively on Y_I and, by adjoints, on X_I . On X_I , P_E has finite rank and retains exactly the coordinates indexed by E . For every $x \in X_I$ and every $\eta > 0$, there is such an E with $\|Q_E x\|_I < \eta$.*

Proof. The nonempty intersections of an incomparable segment family with E are again incomparable segments; the same is true of the portions outside E . Here prefix closure ensures that a segment is cut into an initial part and a terminal part. Applying these restricted tests to the original array proves both contraction inequalities on Y_I , first on finite arrays and then on the completion. The adjoints are contractions and act as the stated coordinate projections on each e_v , hence on X_I by Lemma 9.2. The range of P_E is contained in the span of the finitely many e_v with $v \in E$. Finally, approximate x by a finitely supported vector and let E contain its support and all its ancestors. The contraction bound for Q_E gives the last assertion. $\square$

9.3 Replacing coefficients and controlling cancellation

The following estimate is uniform in the finite prefix and allows an arbitrary displacement. Its tail-supported case is the symmetric splitting estimate for this model.

Theorem 9.5. *Let $E \subset \mathcal{T}$ be finite and prefix closed. If $x, y \in X_I$ satisfy*

$$P_E x = x, \quad \|x + y\|_I \leq 1, \quad \|x - y\|_I \leq 1,$$

then

$$\|Q_E y\|_I \leq (\sqrt{2} + \sqrt{6})\sqrt{1 - \|x\|_I}. \quad (90)$$

In particular, if $Q_E y = y$, the same bound holds for $\|y\|_I$.

Proof. Convexity gives $\|x\|_I \leq 1$. Choose $f \in Y_I$ supported on E such that

$$\|f\|_{Y_I} \leq 1, \quad x(f) = \|x\|_I.$$

Indeed, Hahn–Banach gives a norming element of X_I^* ; reflexivity identifies it with an element of Y_I , and applying P_E preserves its pairing with x without increasing its norm. If $x = 0$, we may simply take $f = 0$.

For an atom with representation (88), define

$$p_i^A = z_{S_i}(f), \quad d(A) = 1 - A(f) = 1 - \sum_i a_i p_i^A \geq 0.$$

Both coefficient vectors (a_i) and (p_i^A) have Euclidean norm at most one. The identity

$$2d(A) = \sum_i (a_i - p_i^A)^2 + \left(1 - \sum_i a_i^2\right) + \left(1 - \sum_i (p_i^A)^2\right)$$

therefore gives

$$\sum_i (a_i - p_i^A)^2 \leq 2d(A), \quad 1 - \sum_i (p_i^A)^2 \leq 2d(A). \quad (91)$$

Replace the coefficients on the terminal pieces by setting

$$\hat{A} = \sum_i p_i^A z_{S_i \setminus E}.$$

Those nonempty terminal pieces remain incomparable. The atom norm bound and (91) imply

$$\|\hat{A} - Q_E A\|_I \leq \sqrt{2d(A)}. \quad (92)$$

Let $\mathcal{R}$ be the minimal nodes of $\mathcal{T} \setminus E$. They form an antichain and their descendant subtrees partition the complement. If $p_i^A \neq 0$ and $S_i \setminus E$ is nonempty, the original segment meets E and exits it through exactly one $q \in \mathcal{R}$. For a fixed q , at most one segment in an incomparable family can do this, since two such segments would both contain q . We may write

$$\hat{A} = \sum_{q \in \mathcal{R}} r_q^A z_{[q, t_q^A]}. \quad (93)$$

Only finitely many coefficients are nonzero. When $r_q^A \neq 0$, let $s_q^A \in E$ be the starting node of the original segment and let $t_q^A \succeq q$ be its terminal node. Then r_q^A is the sum of f from s_q^A to the predecessor of q . For zero coefficients choose $t_q^A = q$; no starting node is needed.

We next show that opposite coefficients in two atoms consume their norming defects:

$$\sum_{q: r_q^A r_q^{A'} < 0} |r_q^A r_q^{A'}| \leq \frac{1}{2} \left(1 - \sum_i (p_i^A)^2 + 1 - \sum_i (p_i^{A'})^2 \right) \leq d(A) + d(A'). \quad (94)$$

For any exit occurring in this sum, the two starting nodes lie on the same ancestor path of q . They are distinct, since identical starts would give identical sums of f , rather than opposite signs. Consider first the exits for which $s_q^A \prec s_q^{A'}$. Shorten the corresponding segment of A to the interval from s_q^A to the predecessor of $s_q^{A'}$. This interval is nonempty and is a subsegment of the original segment. Distinct exits correspond to distinct segments, so all these shortenings can be made simultaneously. Every modified family is still incomparable. Its corresponding squared evaluation changes from $(r_q^A)^2$ to $(r_q^A - r_q^{A'})^2$, an increase of

$$(r_q^{A'})^2 + 2|r_q^A r_q^{A'}| \geq 2|r_q^A r_q^{A'}|.$$

All its squared evaluations sum to at most one. Thus the sum of these products is at most $\frac{1}{2}(1 - \sum_i (p_i^A)^2)$. For the exits with the opposite ordering, shorten the segments of A' instead and obtain the other half of (94). The final inequality follows from (91).

We have controlled the replacement error for one atom and opposite exit coefficients for two atoms. It remains to apply these facts to convex combinations approximating the two endpoints $x + y$ and $x - y$. Fix $\varepsilon > 0$. By Lemma 9.2, choose a finite probability distribution of atoms for each endpoint whose mean differs from that endpoint in norm by at most ε . Choose $\sigma \in \{-1, 1\}$ with equal probabilities, then choose A from the distribution for $x + \sigma y$. Consequently

$$\|\mathbb{E}A - x\|_I \leq \varepsilon, \quad \|\mathbb{E}(\sigma A) - y\|_I \leq \varepsilon, \quad 0 \leq \bar{d} := \mathbb{E}d(A) \leq 1 - \|x\|_I + \varepsilon.$$

Contractivity of Q_E , the identity $Q_E x = 0$, and Cauchy–Schwarz applied to (92) give

$$\|\mathbb{E}\hat{A}\|_I \leq \varepsilon + \sqrt{2\bar{d}}, \quad \|\mathbb{E}(\sigma \hat{A}) - Q_E y\|_I \leq \varepsilon + \sqrt{2\bar{d}}. \quad (95)$$

Put $m_q = \mathbb{E}r_q^A$ and $b_q = \mathbb{E}|r_q^A|$. Apply (94) to independent copies A, A' of the unconditional atom distribution. With $r_+ = \max(r, 0)$ and $r_- = \max(-r, 0)$, independence gives

$$2 \sum_q \mathbb{E}[(r_q^A)_+] \mathbb{E}[(r_q^A)_-] \leq 2\bar{d}.$$

Since $b_q^2 - m_q^2 = 4\mathbb{E}[(r_q^A)_+] \mathbb{E}[(r_q^A)_-]$,

$$\sum_q b_q^2 \leq \sum_q m_q^2 + 4\bar{d}. \quad (96)$$

At this fixed ε , the distributions are finite and their atoms have finite support. Thus all sums of m_q and b_q above are finite, even though $\mathcal{R}$ can be infinite.

An array supported on the antichain $\mathcal{R}$ has Y_I -norm equal to its Euclidean norm: a segment meets at most one of its support nodes, and singleton segments realize the Euclidean test. The coefficient of $\mathbb{E}\hat{A}$ at q is m_q , so duality gives

$$\left(\sum_q m_q^2 \right)^{1/2} \leq \|\mathbb{E}\hat{A}\|_I. \quad (97)$$

For $g \in Y_I$ with $\|g\|_{Y_I} \leq 1$, define

$$L_q = \sup_{t \geq q} |z_{[q,t]}(g)|.$$

For each finite subset of exits, choose segments approaching these suprema. The chosen segments are incomparable, so their squared evaluations sum to at most one. Passing to the suprema and then taking the supremum over finite subsets proves $\sum_q L_q^2 \leq 1$; no supremum need be attained. It follows that

$$|\mathbb{E}(\sigma \hat{A})(g)| \leq \sum_q b_q L_q \leq \left(\sum_q b_q^2 \right)^{1/2}.$$

Taking the supremum over g , and using (95)–(97), yields

$$\|Q_E y\|_I \leq \varepsilon + \sqrt{2\bar{d}} + \left((\varepsilon + \sqrt{2\bar{d}})^2 + 4\bar{d} \right)^{1/2}.$$

The right-hand side is increasing in $\bar{d}$. Substitute $\bar{d} \leq 1 - \|x\|_I + \varepsilon$ and let $\varepsilon \downarrow 0$. The two limiting contributions are $\sqrt{2(1 - \|x\|_I)}$ and $\sqrt{6(1 - \|x\|_I)}$, proving (90). $\square$

9.4 Separated midpoint families

The next corollary removes the finite-prefix condition on the center: if $x \pm u_j$ lie in the unit ball for an infinite separated family (u_j) , then the norm of the common midpoint x is bounded away from one.

Corollary 9.6. *Let $0 < c \leq 1$ and let $x, u_j \in X_I$, $j \in \mathbb{N}$, satisfy*

$$\|x + u_j\|_I, \|x - u_j\|_I \leq 1, \quad \|u_j - u_k\|_I \geq 2c \quad (j \neq k).$$

Then, with $K = \sqrt{2} + \sqrt{6}$,

$$\|x\|_I \leq 1 - (c/K)^2 < 1. \tag{98}$$

Proof. Fix $0 < \eta < c$ and use Lemma 9.4 to choose E with $\|Q_E x\|_I < \eta$. The sequence (u_j) is bounded, since $u_j = ((x + u_j) - (x - u_j))/2$. The range of P_E is finite dimensional, so there are distinct j, k such that, for $w = (u_j - u_k)/2$, one has $\|P_E w\|_I < \eta$. The separation assumption gives $\|w\|_I \geq c$. Moreover, $x + w$ is the midpoint of $x + u_j$ and $x - u_k$, and $x - w$ is the midpoint of the other two endpoints. Hence $\|x \pm w\|_I \leq 1$. Therefore

$$\|P_E x \pm Q_E w\|_I \leq 1 + 2\eta, \quad \|Q_E w\|_I \geq c - \eta.$$

Apply Theorem 9.5 after dividing both vectors by $1 + 2\eta$:

$$\frac{c - \eta}{1 + 2\eta} \leq K \sqrt{1 - \frac{\|P_E x\|_I}{1 + 2\eta}}.$$

Since $\|P_E x - x\|_I < \eta$, letting $\eta \downarrow 0$ gives $c \leq K \sqrt{1 - \|x\|_I}$, which is (98). $\square$

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